



TEAMCENTER

Teamcenter Update Using Deployment Center

Software Version 2606

Unpublished work. © 2026 Siemens

This Documentation contains trade secrets or otherwise confidential information owned by Siemens Industry Software Inc. or its affiliates (collectively, "Siemens"), or its licensors. Access to and use of this Documentation is strictly limited as set forth in Customer's applicable agreement(s) with Siemens. This Documentation may not be copied, distributed, or otherwise disclosed by Customer without the express written permission of Siemens, and may not be used in any way not expressly authorized by Siemens.

This Documentation is for information and instruction purposes. Siemens reserves the right to make changes in specifications and other information contained in this Documentation without prior notice, and the reader should, in all cases, consult Siemens to determine whether any changes have been made.

No representation or other affirmation of fact contained in this publication shall be deemed to be a warranty or give rise to any liability of Siemens whatsoever.

If you have a signed license agreement with Siemens for the product with which this Documentation will be used, your use of this Documentation is subject to the scope of license and the software protection and security provisions of that agreement. If you do not have such a signed license agreement, your use is subject to the Siemens Universal Customer Agreement, which may be viewed at <https://www.sw.siemens.com/en-US/sw-terms/base/uca/>, as supplemented by the product specific terms which may be viewed at <https://www.sw.siemens.com/en-US/sw-terms/supplements/>.

SIEMENS MAKES NO WARRANTY OF ANY KIND WITH REGARD TO THIS DOCUMENTATION INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, AND NON-INFRINGEMENT OF INTELLECTUAL PROPERTY. SIEMENS SHALL NOT BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, CONSEQUENTIAL OR PUNITIVE DAMAGES, LOST DATA OR PROFITS, EVEN IF SUCH DAMAGES WERE FORESEEABLE, ARISING OUT OF OR RELATED TO THIS DOCUMENTATION OR THE INFORMATION CONTAINED IN IT, EVEN IF SIEMENS HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

TRADEMARKS: The trademarks, logos, and service marks (collectively, "Marks") used herein are the property of Siemens or other parties. No one is permitted to use these Marks without the prior written consent of Siemens or the owner of the Marks, as applicable. The use herein of third party Marks is not an attempt to indicate Siemens as a source of a product, but is intended to indicate a product from, or associated with, a particular third party. A list of Siemens' Marks may be viewed at: <https://www.sw.siemens.com/en-US/trademarks/>. The registered trademark Linux® is used pursuant to a sublicense from LMI, the exclusive licensee of Linus Torvalds, owner of the mark on a world-wide basis.

About Siemens Digital Industries Software

Siemens Digital Industries Software is a global leader in the growing field of product lifecycle management (PLM), manufacturing operations management (MOM), and electronic design automation (EDA) software, hardware, and services. Siemens works with more than 100,000 customers, leading the digitalization of their planning and manufacturing processes. At Siemens Digital Industries Software, we blur the boundaries between industry domains by integrating the virtual and physical, hardware and software, design and manufacturing worlds. With the rapid pace of innovation, digitalization is no longer tomorrow's idea. We take what the future promises tomorrow and make it real for our customers today. Where today meets tomorrow. Our culture encourages creativity, welcomes fresh thinking and focuses on growth, so our people, our business, and our customers can achieve their full potential.

Support Center

Log in to Support Center to access software downloads and updates, product and release documentation, Knowledge Base articles, videos, other self-service resources, and to open support cases: <https://support.sw.siemens.com>.

If you do not have a Support Center login, register here: <https://support.sw.siemens.com/register>.

To send feedback on documentation: https://support.sw.siemens.com/doc_feedback_form.

Table of Contents

- Quick Start: Basic Update..... 1-1**
- Quick Start: Platform Update 2-1**
- Plan the update of your Teamcenter environment.....3-1**
 - Where to start.....3-1**
 - Get documentation..... 3-1
 - Get software..... 3-4
 - System requirements..... 3-4
 - Upgrading templates.....3-7**
 - Templates.....3-7
 - How templates are supplied..... 3-8
 - Business Modeler IDE..... 3-8
 - Using the Upgrade Assistant..... 3-9**
 - Download the Upgrade Assistant..... 3-9
 - Install the Upgrade Assistant (Windows)..... 3-9
 - Install the Upgrade Assistant (Linux)..... 3-10
 - Generate an Upgrade Assistant report..... 3-10
 - Using the Instance Data Sync utility..... 3-12**
 - Install the Siemens License Server.....3-21**
 - Upgrading database servers..... 3-23**
 - Supported database server versions..... 3-23
 - Configuring character settings in non-English locales..... 3-23
 - Set shell limits and parameters for Oracle on Linux..... 3-23
 - Upgrade an Oracle server and database..... 3-25
 - Upgrade Microsoft SQL Server databases..... 3-31
- Build a target pre-production environment.....8-1**
 - Prepare the production environment for export.....8-1**
 - Terminate Teamcenter sessions..... 8-1
 - Preserve content security policy directives..... 8-2
 - Back up existing Teamcenter data..... 8-2
 - Clean unused columns from the database..... 8-3
 - Export the production environment..... 8-3**
 - Set default FMS keys..... 8-4
 - Copy the production database..... 8-5
 - Restore FMS keys..... 8-5
 - Copy Teamcenter volume data..... 8-6
 - Export the existing production environment..... 8-6
 - Copy custom extensions..... 8-7
- Build a pre-production environment..... 10-1**

Import the copied environment into the pre-production environment.....	10-1
Install a preproduction Deployment Center server.....	10-1
Import the copy of the production database.....	10-1
Import the copied environment into Deployment Center.....	10-2
Update the preproduction environment.....	11-1
Prepare the pre-production environment for update.....	11-1
Install C Shell on Linux servers.....	11-1
Verify required groups and roles.....	11-1
Verify required character set.....	11-2
Update the pre-production environment.....	11-3
Upgrade Deployment Center.....	11-3
Migrate BMIDE custom templates.....	11-3
Update the Software Repository.....	11-3
Update the pre-production environment.....	11-4
Post-update tasks.....	11-4
Run the postinstallation tasks script (Linux systems).....	11-4
Enable a UTF-8 database.....	11-5
Configure units of measure.....	11-5
Resolve Schedule Manager model errors.....	11-5
Upgrade Zugriffs-Management rules.....	11-7
Maintain default scope rules.....	11-7
Set the time zone.....	11-8
Update Active Workspace components.....	11-8
Additional postupgrade steps.....	11-11
Deploy the update to your production environment.....	14-1
Update volume servers.....	14-1
Update the production environment.....	14-1
Complete the update.....	16-1
Apply updates to your environment.....	16-1
Import a configuration from TEM.....	16-1
Perform updates or patches.....	16-1
Appendices.....	17-1
Troubleshooting.....	17-1
Troubleshooting Microsoft SQL Server.....	17-1
Troubleshooting Teamcenter upgrade.....	17-2

1. Quick Start: Basic Update

This *Quick Start* guide helps you understand the process of a *basic update* of a Teamcenter environment.

Teamcenter is a leader in product lifecycle management (PLM) software that optimizes how you plan, develop, and deliver products. It is fully configurable and scalable to the needs of small businesses as well as globally distributed organizations. This *Quick Start* guide will familiarize you with the essential steps for updating an existing Teamcenter environment to a new release, with changes that do not require new Teamcenter home and data directories, that is, <TC_ROOT> and <TC_DATA>. These essential steps guide you to detailed topics and related resources on Siemens Support Center.

If you are updating from a version of Teamcenter 2512 or later, you can update your Teamcenter environment to Teamcenter 2606 using this *Basic Update Quick Start* guide. If your existing Teamcenter environment is a version earlier than Teamcenter 2512, use the [Platform Update Quick Start](#) guide.

1
PLAN

Plan the update of your Teamcenter environment

Software planning

1. See the patch release **README.pdf** document for specific information related to the update. Also, see the patch **README.csv** document for defects and PRs that are resolved in the release. These documents are in the [Downloads](#) area on Support Center.
2. See the *Teamcenter-Versionshinweise* for the latest information about issues that may affect your update.

Downloading software

1. Download Teamcenter software from Support Center.
In the Teamcenter [Downloads](#) area on Support Center, choose **Select a Version > Teamcenter 2606 > Teamcenter 2606**, and then download the software kit for your platform:
 - Windows: **Tc2606_wntx64.zip**
 - Linux: **Tc2606_Inx64.zip**Expand the software kit to a local directory.
2. Find additional Siemens and third-party software required for your Teamcenter environment on Support Center, or on the respective third-party vendor site. Review the software and installation requirements for those products. Download the additional Siemens and third-party software you need for your Teamcenter environment.

Infrastructure planning

If you intend to deploy with a *blue-green* approach to minimize downtime, plan for the preproduction environments that include the future Green production environment and an intermediate upgrade server. Learn how the [Instance Data Sync utility](#) supports this approach.

2 EXPORT

Export a copy of the production environment

1. Set FMS keys to their default values.
2. Export the database from the current, or source, production environment. Perform a database dump using the appropriate method for your database type, for example, the **expdp** utility for Oracle.

Note

 - Ensure that no users are connected to the database while you export the database. This step requires downtime to prevent any disruptions or data inconsistencies.
 - After you complete the database export, ensure the database remains in read-only mode to prevent any updates. This step is critical to maintain data integrity and consistency during the migration process.
3. Restore FMS keys to their previous values.
4. Back up secrets in the Teamcenter Secrets Manager, also called Teamcenter Vault.
5. Copy a representative set of Teamcenter volumes for testing. Copy the default volume and one other volume.
6. Export the production environment to an XML configuration, also known as a Quick Deploy file.
7. If you are using a blue-green deployment with temporary parallel production environments with the Instance Data Sync utility, see [Using the Instance Data Sync utility](#) to initiate the tracking of production data changes.

3 IMPORT

Import the production server clone into the intermediate preproduction server environment

1. On your designated preproduction environment machine, install the prerequisite software and settings as you would for a new installation. Install the same software versions as in your production environment.
2. [Install a preproduction Deployment Center server](#) of the same version that you used to export your production environment.
3. Add the following software to the software repository:
 - The Teamcenter software kit for the version used in your production environment.
 - Your custom Teamcenter software from your production environment.
4. [Import the copy of the production database](#).
5. If you intend to deploy using a blue-green approach with the [Instance Data Sync](#) utility, run the utility in the current Green environment to clear any prior database tracking triggers.

6. **Import the copied environment into Deployment Center.**
7. In Deployment Center, change the machine names and the OS values for a simplified Teamcenter environment. Update the properties for all components as needed to reflect the preproduction environment.
8. In the database creation settings, select **Copy Environment using existing populated database**.

Note

This option creates a test environment that cannot be used for production purposes. If you intend to use a blue-green deployment approach with temporary parallel production environments, follow the steps in [Quick Start: Platform Update](#) for creating the production clone.

9. To simplify the preproduction environment, remove extra components added for scaling, such as extra server managers, Dispatcher modules, and microservice nodes.
10. Generate deploy scripts for machines defined in your environment.
11. Run the deployment scripts on server machines in your preproduction environment.
12. Verify access to services and components in the pre-production environment, as when you installed Teamcenter.

Also, verify business objects and administration data are present, such as the preferences, the rule tree, the stylesheets, and so on.



Update the preproduction environment

1. **Prepare the preproduction server for update**
 - a. Install or update prerequisite software to certified versions in the Platform and Software Certification Matrix.
 - b. Upgrade the preproduction Deployment Center server to the latest version.
2. **Update Teamcenter components**
 - a. Add the Teamcenter 2606 software kit, your custom BMIDE template package, and any other software kits in your source environment to the software repository in the preproduction environment.
 - b. In the **Software** tab in Deployment Center, add the Teamcenter 2606 software kit and your custom BMIDE template package to your preproduction environment.
3. **Configure Teamcenter and build deploy scripts**
4. **Deploy updates to the preproduction environment**
 - a. Copy deployment scripts to their specific machines. Run the deployment scripts on server machines in the recommended sequence.

See the *Deployment Sequence (at deployment time)* topic in the Teamcenter-Bereitstellungsreferenz-Architektur for guidance about the order of deployment.

- b. On Linux machines, deploy Teamcenter microservices on Docker Swarm or Kubernetes.

5. Configure certificates

- a. Import certificates from your certificate authority (CA), which may be the Teamcenter Vault, your company's private CA, or a third-party CA.
- b. If you use a private CA or a third-party CA, provide certificates for the complete CA chain to the Teamcenter Vault.
- c. Place each machine-specific certificate in the corresponding machine's Teamcenter home directory (<TC_ROOT>).

6. Add administrative client components

Install administrative client components using your machine-specific deploy scripts:

- Administrative two-tier Rich Client
- BMIDE client
- Teamcenter Management Console.

7. Perform [post-update tasks](#).

8. Verify that essential Teamcenter services and components are operating in the preproduction environment:

Teamcenter Vault

Verify that the Vault Service is in a running state.

Dateiverwaltungssystem (FMS)

Access the FMS ping page:

`http(s)://<fms-master>:4544/Ping`

Deployment is successful if the page displays the FSC ID.

Teamcenter Management Console (for Java EE web servers)

Verify that the console is reachable at:

`http(s)://<console-server>:8083/mgmt/console`

Teamcenter Server Manager

Use Teamcenter Management Console to verify that the Server Manager pools you configured are running.

Active Workspace Gateway

Access the Gateway ping page at:

`http(s)://<gateway-server>:3000/ping`

This page lists the statuses of several diagnostic checks.

Microservice node

	Verify that the Service Dispatcher endpoint, that is, Eureka microservice, is reachable at: http(s)://<node-server>:8787/eureka/v2/apps Active Workspace Indexer Ausführen eines beispielhaften Suchindizierungsablaufs. Teamcenter Dispatcher Verify the Dispatcher installation.
	Deploy the update to your production environment 1. Deploy updates to the production environment a. Copy deployment scripts to their specific machines. Run the deployment scripts on server machines in the same sequence that you used in the preproduction environment. See <i>Deployment Sequence (at deployment time)</i> in the Teamcenter-Bereitstellungsreferenz-Architektur for guidance about order of deployment. b. On Linux machines, deploy Teamcenter microservices on Docker Swarm or Kubernetes. c. Merge your custom extensions as you did with your preproduction environment. d. Import your certificates as you did in your preproduction environment.
	Complete the update 1. Verify access to Teamcenter services and components as you did in your preproduction environment. 2. Log on to Active Workspace and the administrative two-tier Rich Client to test user authentication and basic functionality. 3. After you validate the updated production environment, run mass client deploy scripts to install the Rich Client and other client components on all the client machines in your production environment. 4. Log on to Teamcenter from a non-administrative client machine to ensure client functionality for all users. 5. If you are using a blue-green deployment with temporary parallel production environments, see Using the Instance Data Sync utility to sync the production data changes to your Green environment.

For *Quick Start* guides for other Teamcenter deployment processes, see the *Teamcenter Deployment Guidemap*.

2. Quick Start: Platform Update

This *Quick Start* guide helps you understand the process of a *platform update* for a Teamcenter environment.

Teamcenter is a leader in product lifecycle management (PLM) software that optimizes how you plan, develop, and deliver products. It is fully configurable and scalable to the needs of small businesses, as well as globally distributed organizations. Reviewing this *Quick Start* guide will familiarize you with the essential steps for updating an existing Teamcenter environment to a new release that contains significant changes to the platform requirements of your current release. These essential steps guide you to detailed topics and related resources on Siemens Support Center.

This *Platform Update* is based on the industry *blue-green* deployment approach, using intermediate servers to maximize uptime during the update process.

Teamcenter 2606 supports direct update from Teamcenter 14.0 or later. If your current version is earlier than 14.0, you must update to version 14.0 or later *before* you update to Teamcenter 2606.

If you are updating from version 14.0 or later, you can update to Teamcenter 2606 using this *Quick Start: Platform Update* guide.

If you are updating Teamcenter 2606 with a Teamcenter 2606.yyymm update patch, use the [Quick Start: Basic Update](#) guide.

Note

For complete supported update paths, see the Teamcenter Interoperability Matrix ([Teamcenter Interoperability Matrix <date>.xlsx](#)) on the [Support White Papers Certifications](#) page on Support Center.

1
PLAN

Plan the update of your Teamcenter environment

Software planning

1. See *What's New in Teamcenter 2606* for information about the new features in this release.

Also, see the *What's New* documents for the other versions of Teamcenter between your currently-installed version and Teamcenter 2606 to be aware of other new features that may affect your environment.
2. See the *Teamcenter-Versionshinweise* for the latest information about issues that may affect your update.
3. See the Teamcenter 2606 **README.csv** document for defects and problem reports (PRs) that are resolved in the release. This document is in the [Downloads](#) area on Support Center.

4. Many Teamcenter applications are described in dedicated *Deployment and Administration* guides. To locate these guides for the applications in your environment, see the 1. Halt: Teamcenter-Dokumentationsstartseite page.
5. Informationen zu Betriebssystemen und Software von externen Anbietern, die für Teamcenter-Komponenten in dieser Version zertifiziert ist, finden Sie in der Teamcenter 2606-Plattform- und Software-Zertifizierungsmatrix. Wählen Sie im Bereich Teamcenter **Downloads** die Option **Version auswählen** > **Support White Papers Certifications** aus und laden Sie dann **Tc 2606 PlatformMatrix- Datum .xlsx** herunter.
6. If you use commercial software integrations with Teamcenter, find Teamcenter 2606-compatible versions of those integrations from the **Product Compatibility Matrices** tile in the **Support Center** home page.
7. Plan for potential impacts to your custom extensions to Teamcenter.
 - a. Review *BMIDE for Data Model Design* for steps to ensure your BMIDE templates are compatible with the latest version of Teamcenter.
 - b. See the *Änderungen in Teamcenter-APIs* reference on Support Center to learn about changes to deprecated or obsolete APIs and plan for refactoring impacted code.
 - c. Use the **Upgrade Assistant ITK Reporter** tool to identify deprecated and obsolete APIs before you begin updating your Teamcenter customizations.

Infrastructure planning

1. Download the **Teamcenter Deployment Reference Architecture** ZIP package from the Teamcenter **Downloads** area on Support Center.
2. From the *Teamcenter Deployment Reference Architecture* document, learn about environment types and components added to or removed from the Teamcenter architecture. Determine which machines in your environment may need operating system updates, and then plan a *preproduction environment* architecture that reflects your current production environment and allows for changes in the latest version.
3. If you intend to deploy with a *blue-green* approach to minimize downtime, plan for preproduction environments that include the future Green production environment and an intermediate upgrade server. Learn how the **Instance Data Sync utility** supports this approach.

Downloading software

1. Download Teamcenter software from Support Center.

In the Teamcenter **Downloads** area on Support Center, choose **Select a Version** > **Teamcenter 2606** > **Teamcenter 2606**, and then download the software kit for your platform:

- Windows: *Tc2606_wntx64.zip*
- Linux: *Tc2606_lnx64.zip*

Expand the software kit to a local directory.

2. Find additional Siemens and third-party software required for your Teamcenter environment on Support Center, or on the respective third-party vendor site. Review the software and installation requirements for those products. Download the additional Siemens and third-party software you need for your Teamcenter environment.

2 CREATE

Create a new target Teamcenter 2606 production environment.

1. install the prerequisite software and settings for a new Teamcenter installation.
2. Install Deployment Center 2606.
3. Install the BMIDE standalone component with which to migrate custom templates to Teamcenter 2606.
4. Create a new Teamcenter production environment using Teamcenter 2606 with the same custom templates installed in your source Teamcenter environment.
5. Update the site ID of this new target environment to match the site ID of the source production environment.
6. Install all the same applications and components that are installed in your source production environment. At this time, do not install multiple instances of components like FSC or Active Workspace Gateway. You can add these extra components to scale the environment after the update is complete.

Note

If necessary, review the `<TC_DATA>\model\master.xml` file to get the list of installed templates in the source Teamcenter version.

7. Make sure that the passwords used in this environment match those from existing Deployment Center instances. Passwords need to match because database credentials, FMS keystore passwords, and so on, are now stored in the Deployment Center Vault.
8. Conduct User Acceptance Testing (UAT) on the new Teamcenter 2606 environment to ensure the installation is valid.

3 EXPORT

Export the current production environment configuration and database

You will use these to create an intermediate preproduction server environment.

1. On your designated preproduction environment machine, install the prerequisite software and settings as you would for a new installation. Install the same software versions as in your production environment.
2. Export the source environment using Quick Deploy by running the following command:

```
dc_quick_deploy -mode=export -dcurl=<dc-url>  
-environment=<source-DC-environment-name>  
-exportType=Full -exportfile=<export-file-name>  
-dcusername=<dc-user> -dcpassword=<dc-password>
```

3. Modify the exported Quick Deploy XML file to create a *single-box environment*, keeping only a minimal set of components, such as Corporate Server, FSC, Database Server, License Server, FSC Group, FSC Keys, and any other required components like Teamcenter Vault and HTTPS Config. Update the hostnames of the components to reflect the intermediate server.

To simplify the preproduction environment, remove any extra components added for scaling, such as extra server managers, Dispatcher modules, and microservice nodes.

4. Export the database from the current, or source, production environment. Perform a database dump using the appropriate method for your database type, for example, the **expdp** utility for Oracle.

Note

- Ensure that no users are connected to the database while you export the database. This step requires downtime to prevent any disruptions or data inconsistencies.
- After you complete the database export, ensure the database remains in read-only mode to prevent any updates. This step is critical to maintain data integrity and consistency during the migration process.
- If you intend to use the **Instance Data Sync utility**, run the utility in the current Blue environment to track database changes.

4 IMPORT

Import the production server clone into the intermediate preproduction server environment

1. On your designated preproduction environment machine, install the prerequisite software and settings as you would for a new installation. Install the same software versions as in your production environment.
2. Install the version of Deployment Center you use on your current, or source, production server. Apply any migration hotfixes to the Deployment Center instance.
3. Import the modified Quick Deploy file from the source environment into the new intermediate preproduction environment, keeping the same environment name and site ID. This creates a single-box replica of the existing source environment.

```
dc_quick_deploy -mode=export -dcurl=<dc-url>  
-environment=<source-DC-environment-name>  
-inputFile=<export-file-name> -dcusername=<dc-user>  
-dcpassword=<dc-password>
```

If the **-importenvironmentmetadata** argument is not supported with the current Deployment Center version, update the environment site ID in the Deployment Center web portal, in the **Environment > Overview** page. This lets you keep the same site ID as the source environment.

4. Generate deployment scripts and copy them to their target machines. Run the deployment scripts to create a single-box corporate server environment on this intermediate server with all the data model templates.
5. Stop all Teamcenter services including the Read Expression Manager (**am_read_expression_manager**) and Revision Configuration Accelerator (**revision_config_accelerator**) in the intermediate environment.
6. Delete the existing database user and tablespaces in the intermediate environment DB.
7. Import the dumped database into the intermediate Teamcenter environment using the appropriate tool for your database type, for example, **impdp** for Oracle.
8. Start all Teamcenter services in the intermediate environment.
9. Copy and replace the **symmetric_key_keystore.jceks** from the production environment to the intermediate environment `<TC_ROOT>\fsc\fsc_config` folder, or fuse the following steps to update the FSC key in the database to match the keystore generated during the new deployment. You can also update the FSC keys configuration (`<TC_ROOT>\fsc\fsc_config\symmetric_key_keystore.jceks`) on the new target environment to match that of the intermediate upgraded environment.
 - a. Download **Keystore Explorer** or a similar tool to retrieve the key from the keystore. (Skip this step if you use a different tool.)
 - b. In your keystore tool, open the following existing keystore:
`<TC_ROOT>\fsc\fsc_config<keystore>.jceks`
 - c. Enter the password you provided for the **Keystore Password** property in the **FSC Keys** component in Deployment Center when you installed the intermediate Teamcenter environment. This will display the alias information.
 - d. Click the alias and enter the keystore password. This displays the alias information.
 - e. Copy the key displayed in the **Encoded** field.
 - f. From the Teamcenter command shell, run the following command:

```
install_encryptionkeys.exe -u=<Tc-admin-user> -p=<Tc-admin-password> -f=list
```

This command displays the existing key in the database.

- g. If these keys do not match, you must update the key in the database. From the same Teamcenter command shell, run the following command:

```
install_encryptionkeys.exe -u=<Tc-admin-user> -p=<Tc-admin-password>  
-f=modify -key=<key-retrieved-from-FSC-keystore-file>
```

This command updates the key in the database.

- h. Verify the key by rerunning the utility with the **-f=list** option.

10. If you intend to use the **Instance Data Sync utility**, run the utility in the current Green environment to clear any prior database tracking triggers.
11. Synchronize volumes by copying the volumes from the production environment in the intermediate environment machine. This step is not required if you use a shared volume.
12. Read the volume ID of the production environment within `<TC_ROOT>\fsc\backup.xml -> volumeUiD -> volumeUid`.
13. Update the production environment volume ID and volume path in the intermediate environment FMS master XML.
14. Restart the FSC service.
15. Verify access to services and components in the preproduction environment, as when you installed Teamcenter.

Also, verify that business objects and administration data are present, such as preferences, the rule tree, stylesheets, and so on.
16. Verify that essential Teamcenter services and components are operating in the preproduction environment:

Teamcenter Vault

Verify that the Vault Service is in a running state.

Dateiverwaltungssystem (FMS)

Access the FMS ping page:

`http(s)://<fms-master>:4544/Ping`

Deployment is successful if the page displays the FSC ID.

Teamcenter Management Console

Verify that the console is reachable at:

`http(s)://<console-server>:8083/mgmt/console`

Teamcenter Server Manager

Use Teamcenter Management Console to verify that the Server Manager pools you configured are running.

Active Workspace Gateway

Access the Gateway ping page at:

`http(s)://<gateway-server>:3000/ping`

This page lists the statuses of several diagnostic checks.

Microservice Node

Verify that the Service Dispatcher endpoint, that is, Eureka microservice, is reachable at:

`http(s)://<node-server>:8787/eureka/v2/apps`

	Active Workspace Indexer Ausführen eines beispielhaften Suchindizierungsablaufs. Teamcenter Dispatcher Verify the Dispatcher installation.
5 UPDATE	Update the intermediate cloned environment to the target Teamcenter version Prepare the preproduction server for the update - Install or update the prerequisite software to the certified versions in the Platform and Software Certification Matrix. - Upgrade the preproduction Deployment Center server to the latest version. - Update the cloned intermediate environment to Teamcenter 2606 without adding any additional templates. - Upgrade Deployment Center on the intermediate server to Deployment Center 2606. Update your custom BMIDE template. - Install a standalone Business Modeler IDE client for the preproduction environment. - Upgrade the template project to the current data model format. - Create and deploy an updated BMIDE package. Update extensions - Copy your custom extensions from your source control system or from the Teamcenter environment. - Refactor for obsolete APIs. - Rebuild your API-based extensions based on the certified compilers listed in the Platform and Software Certification Matrix. - Merge file-based changes with new default files. - Migrate or de-customize extensions to leverage new target capabilities in Teamcenter 2606. - Merge custom Active Workspace stylesheets, the rule tree, and other administration data with the new default templates. Update Teamcenter components - Add the Teamcenter 2606 software kit, your custom BMIDE template package, and any other software kits in your source environment to the software repository. Note When you add the latest software kit to your environment, Deployment Center automatically recognizes whether the new version contains the significant changes that denote a <i>platform update</i>.

When updating to a version with such significant changes, Deployment Center creates a new Teamcenter home directory (<TC_ROOT>) and data directory (<TC_DATA>). You can rename these directories before you deploy the update.

- b. In the **Software** tab in Deployment Center, add the Teamcenter 2606 software kit and your updated custom BMIDE template package to your preproduction environment.
4. Compare the applications with the existing Deployment Center 2606 production environment to ensure the intermediate environment is not missing any features.
5. **Deploy updates to the preproduction environment**
 - a. Copy deployment scripts to their specific machines. Run the deployment scripts on server machines in the recommended sequence.

See the *Deployment Sequence (at deployment time)* topic in the Teamcenter-Bereitstellungsreferenz-Architektur for guidance about the deployment order.
 - b. On Linux machines, deploy Teamcenter microservices on Docker Swarm or Kubernetes.
6. **Configure certificates**
 - a. Import certificates from your certificate authority (CA), which may be the Teamcenter Vault, your company's private CA, or a third-party CA.
 - b. If you use a private CA or third-party CA, provide certificates for the complete CA chain to the Teamcenter Vault.
 - c. Place each machine-specific certificate in the corresponding machine's Teamcenter home directory, that is, <TC_ROOT>.
7. **Add administrative client components**

Install administrative client components using your machine-specific deploy scripts:

 - Administrative two-tier Rich Client
 - BMIDE client
 - Teamcenter Management Console.
8. Perform **post-update tasks**.
9. Conduct UAT testing on the updated environment to ensure everything is installed successfully.

Connect the intermediate cloned environment to the new target environment

1. Create a Teamcenter database backup by exporting the database of the updated cloned environment again, using the same database vendor-specific approach.
2. Stop all Teamcenter services before importing the database.
3. Delete the database user and tablespaces, and import the export of the database in the new target Teamcenter 2606 production environment.

	4. Sync volumes by copying the volumes from the upgraded intermediate environment into the target environment machine. This step is not required if you use a shared volume. 5. Update the FMS Master file to change the volume path and ID. These details can be found in the intermediate updated server, in the <code><fsc_installation_path>\backup.xml</code> file. 6. After restoring the database, FSC keys may malfunction due to a mismatch. To resolve this, either update the FSC key in the database to match the keystore generated during the new deployment or update the FSC keys configuration (<code><TC_ROOT>\fsc\fsc_config\symmetric_key_keystore.jceks</code>) as described earlier in this Quick Start guide. 7. Update the Fms_BootStrap_Urls preference to ensure the correct FSC URL is updated. 8. Update the AWS_FullTextSearch_Solr_URL preference to ensure the correct Solr URL is updated. 9. Restart all the Teamcenter services. 10. Test the environment to verify access to services and components, as you did with the preproduction environment.
7 COMPLETE	Complete the update 1. Conduct basic UAT testing to ensure the environment is working properly before you direct users to the new production environment. 2. Log on to Active Workspace and the administrative two-tier Rich Client to test user authentication and basic functionality. 3. After you validate the updated production environment, run mass client deploy scripts to install the Rich Client and other client components on all the client machines in your production environment. 4. Log on to Teamcenter from a non-administrative client machine to ensure the client functionality for all users. 5. If you are using the Instance Data Sync utility, see Using the Instance Data Sync utility to sync the production data changes to your Green environment.

For *Quick Start* guides for other Teamcenter deployment processes, see the *Teamcenter Deployment Guidemap*.

3. Plan the update of your Teamcenter environment

Plan your Teamcenter update in consideration of all software components in your environment, including infrastructure components like database servers and license server, which may need upgrading before you perform the Teamcenter update.

Plan to perform an update on a pre-production environment to identify and resolve potential issues before you update production systems.

Where to start

Get documentation

Teamcenter documentation is available from two sources:

- **Internet: Support Center**

This is Siemens Digital Industries Software's comprehensive support portal, which provides documentation for all Siemens software products and versions.

You require a Webkey account to access Support Center. However, you can avoid this requirement by installing the *Siemens Documentation Proxy*, which provides secure documentation access using a personalized API key, with no need to log on. Teamcenter clients can be configured to access help through the Documentation Proxy.

- **Intranet: Siemens Documentation Server**

This is a locally installed server that can host documentation for all your Siemens Digital Industries Software products. No Internet access is required. You can configure the server for single-machine use or network-wide access with no Webkey or API key required.

Teamcenter clients can be configured to access the help on the Siemens Documentation Server.

For an orientation to Support Center, see Siemens Software [Support Center videos](#) on YouTube.

Install the Documentation Proxy or the Documentation Server

Log on to Support Center and open the [Siemens Documentation Server Downloads](#) page:

Products→Siemens Documentation Server→Downloads

Choose how you want to access documentation, and then install the Documentation Proxy or the Documentation Server.

Installing Siemens Documentation Proxy	Installing Siemens Documentation Server
- Under Select a Version, choose Documentation Proxy 3, and then click the tile for the latest Documentation Proxy 3.<x> release. - Download the Documentation Proxy installer: Windows: DocumentationProxy.<version>.exe Linux: DocumentationProxy.<version>.aol - Install the Documentation Proxy according to the <i>Documentation Proxy Installation Guide</i> for Windows or Linux, available under Release Documentation on the software download page. Installing the Documentation Proxy requires generating an API key at the Siemens Support Center account site. This may require you to obtain your Siemens site ID from your Teamcenter administrator.	- Under Select a Version, choose Siemens Documentation Server 3, and then click the tile for the latest Siemens Documentation Server 3.<x> release. - Download the Documentation Server installer: Windows: HelpServer.<version>.exe Linux: HelpServer.<version>.aol - Install the Documentation Proxy according to the <i>Siemens Documentation Server Installation Guide</i> for Windows or Linux, available under Release Documentation on the software download page.

Note the machine and port on which you configured the Documentation Proxy or Documentation Server. These are required to configure help access from Teamcenter clients.

Install the Teamcenter 2606 documentation kit

If you installed the Siemens Documentation Proxy, skip this section.

Teamcenter documentation is delivered in *documentation kits*. Each kit contains documentation content and an installation wizard that automatically installs documentation onto your Documentation Server.

- Log on to Support Center and open the [Teamcenter Downloads](#) page:

Products→Teamcenter→Downloads

- Under **Select a Version**, choose **Teamcenter 2606**, and then click the **Teamcenter 2606** tile.
- Install the Teamcenter 2606 documentation onto the Documentation Server:
Windows: Double-click the **docs-teamcenter-2606-<locale>.exe** file.

Linux: Enter the following commands:

```
sudo chmod 777 docs-teamcenter-2606-<locale>.aol
sudo teamcenter-2606-<locale>.aol
```

These commands require administrative privileges.

For more information about installing documentation kits and managing the Documentation Server, see the *Siemens Documentation Server Installation Guide* for Windows or Linux.

Verify documentation access

Open the Teamcenter 2606 documentation from your preferred source:

- Support Center (Webkey logon):

<https://docs.sw.siemens.com/de-DE/doc/282219420/PL20251212545240207.home>

- Support Center (via Documentation Proxy):

<http://<doc-proxy-host>:<doc-proxy-port>/de-DE/doc/282219420/PL20251212545240207.home>

- Siemens Documentation Server:

<http://<doc-server-host>:<doc-server-port>/de-DE/doc/282219420/PL20251212545240207.home>

Enable help access in Teamcenter clients

Configure help in the Rich Client

If you use the Rich Client, configure the **Help** button in the client to open Teamcenter help from your preferred source.

During installation:

When prompted in Deployment Center, enter your preferred documentation URL in the **Documentation server URL** box.

After installation:

Configure help access in the Rich Client as described in the appropriate Rich Client installation guide for Windows or Linux.

Configure help in Active Workspace

Configure the **Help** button in the client to open Teamcenter help from your preferred source.

During installation:

The Active Workspace **Help** button links to Support Center by default and cannot be changed during installation. Accessing help directly on Support Center requires a Webkey account.

After installation:

If you use the Documentation Proxy or the Documentation Server, set the **TC_Help_Documentation_Link** preference to the path to your preferred documentation source. This configures the Active Workspace **Help** button to link to that source.

If you use Support Center via the Documentation Proxy, set this preference to **http://<doc-proxy-host>:<doc-proxy-port>/de-DE/doc/282219420/PL20251212545240207**.

If you use Siemens Documentation Server, set the preference to **http://<doc-server-host>:<doc-server-port>/de-DE/doc/282219420/PL20251212545240207**.

Get software

Updating Teamcenter requires the Teamcenter software kit, which includes microservice framework and Active Workspace software.

1. Log on to Support Center and open the [Teamcenter Downloads](#) page:

Products→Teamcenter→Downloads

2. Under **Select a Version**, choose **Teamcenter 2606**, and then click the **Teamcenter 2606** tile.
3. Download the Teamcenter 2606 software kit for your platform:
 - Windows: **Tc2606_wntx64.zip**
 - Linux: **Tc2606_inx64.zip**
4. Unzip its contents to a local directory.

If an update (patch) to Teamcenter 2606 is available, for example, Teamcenter 2606.2608, you can additionally download the update, and apply it during the Teamcenter update.

Can I place the software in a remote location?

Your primary repository in Deployment Center must be a local path. However, you can specify additional repository locations, and these may be UNC paths or local file system paths. Mapped drives are not supported for any software repositories in Deployment Center. For more information, see *Deployment Center — Verwendung*.

System requirements

Verify system software requirements

1. Log on to Support Center and open the [Support White Papers Certifications](#) page:
 - a. Open **Products→Teamcenter→Downloads**.

- b. Under **Select a Version**, choose **Support White Papers**→**Support White Papers Certifications**, and then click the **Support White Papers Certifications** tile.
2. Download the following support documents:

Software Certifications Matrix (Tc2606PlatformMatrix-<date>.xlsx)

Contains information about system software certified for Teamcenter, such as operating systems and Java runtime environments (JREs).

Teamcenter Interoperability Matrix (Teamcenter Interoperability Matrix <date>.xlsx).

Lists supported Teamcenter versions that can be updated to Teamcenter 2606. Also lists versions of Siemens Digital Industries Software products that are compatible with Teamcenter 2606.

The Teamcenter Interoperability Matrix also correlates versions of Deployment Center with compatible versions of Teamcenter, and shows supported paths for upgrading Deployment Center. For information about upgrading Deployment Center, see *Deployment Center — Verwendung*.

Make sure you install versions of the required software that are listed in the Software Certifications Matrix and the Teamcenter Interoperability Matrix.

Supported updates

See the Teamcenter Interoperability Matrix for versions of Teamcenter that can be updated to Teamcenter 2606.

If your current Teamcenter version is earlier than those supported for update, you must update to a supported version before you update to Teamcenter 2606.

Platforms

Teamcenter 2606 is certified on the following platforms:

- **On-premises**
 - Microsoft Windows Server (server components only)
 - Microsoft Windows Desktop (client components only)
 - SUSE Linux
 - Red Hat Linux
- **Cloud**
 - Amazon Web Services
 - Microsoft Azure
 - SUSE Linux
 - Google Cloud Platform

Informationen zu Betriebssystemen und Software von externen Anbietern, die für Teamcenter-Komponenten in dieser Version zertifiziert ist, finden Sie in der Teamcenter 2606-Plattform- und Software-

Zertifizierungsmatrix. Wählen Sie im Bereich Teamcenter **Downloads** die Option **Version auswählen > Support White Papers Certifications** aus und laden Sie dann **Tc 2606 PlatformMatrix- Datum .xlsx** herunter.

The Platform and Software Certification Matrix contains a tab for on-premise platforms and separate tabs for each certified cloud platform. See the appropriate tab for your platform type.

Note the following additional considerations for certain platform types:

Microsoft Windows

- On Windows platforms, disable Windows User Account Control (UAC) before you install Teamcenter. This option is available in the **Control Panel→User Accounts** dialog box.

Windows UAC can interfere with Teamcenter installation programs. Siemens Digital Industries Software recommends turning off UAC for administrative users only.

For more information, see Microsoft Windows documentation.

- If you use a non-English language operating system version of Windows, you must install and enable the Multilingual User Interface (MUI) pack to ensure the language font is displayed properly.
 1. Download and install the MUI pack for Windows from Microsoft.
 2. Open the **Regional and Language Options** dialog box in the Windows Control Panel.
 3. In the **Languages** tab, set the required language for the menus and dialogs.
 4. In the **Advanced** tab and the **Regional Options** tab, set the required language.

Linux

- Linux hosts must have graphics capabilities to run Teamcenter installation tools.

For operating system requirements, see the Hardware and Software Certifications knowledge base article on Support Center.
- Linux hosts must have the **nslookup** utility available to ensure operation of the license server.
- Make sure Linux host names do not exceed 31 characters in length. Host names longer than 31 characters cause Teamcenter corporate server installation to fail during saving of the POM schema file in the <TC_DATA> directory.

Teamcenter installation tools do not require fully qualified domain names for host names. If your fully qualified domain name exceeds 31 characters, use the server short host name instead.

For more information, see the solutions document 002-7004480 on Support Center.

- Teamcenter Environment Manager (TEM) requires the ISO8859-1 character set. Make sure this character set is available on your host.

Database

Teamcenter requires a relational database management system (RDBMS) for storing Teamcenter data. Before you install Teamcenter, you must install an Oracle database server or a Microsoft SQL Server database server.

If your database server is not a supported version, [upgrade your database server](#) to a supported version before you install Teamcenter.

Choose a database management system that suits the platforms of your Teamcenter servers and clients, and make sure your Teamcenter corporate server host has access to the database server.

If you use Oracle, [set system parameters](#) to recommended values to ensure adequate database performance.

Web browser

A web browser is required if you use the following:

- Teamcenter online help
- Active Workspace
- Deployment Center

For these products, Teamcenter supports the following web browsers:

- Windows systems: Microsoft Edge, Mozilla Firefox, and Google Chrome
- Linux systems: Mozilla Firefox and Google Chrome

Informationen zu Betriebssystemen und Software von externen Anbietern, die für Teamcenter-Komponenten in dieser Version zertifiziert ist, finden Sie in der Teamcenter 2606-Plattform- und Software-Zertifizierungsmatrix. Wählen Sie im Bereich Teamcenter **Downloads** die Option **Version auswählen** > [Support White Papers Certifications](#) aus und laden Sie dann **Tc 2606 PlatformMatrix- Datum .xlsx** herunter.

Upgrading templates

Templates

A *template* is a container that holds data model definitions. A template can contain any number of business objects, classes, lists of values, and business rules. Any Teamcenter feature that adds definitions to the data model has an associated template. In addition, templates can be supplied to you from another Teamcenter site, partner, or third party.

Achtung

You *must* have access to all templates used at your site before beginning the upgrade of the corporate server. This is very important. During the corporate server upgrade, you load all templates used by your site. Templates can only be loaded during the upgrade of the corporate server. They cannot be loaded after the upgrade. Failing to load all templates your site depends on causes problems in managing the data model and can result in loss of data, time, and money.

How templates are supplied

Templates are supplied by:

- The Teamcenter software kit
- Software kits for asynchronous Teamcenter releases and integrations
- Other Teamcenter sites, partners, or third parties

Templates supplied on the Teamcenter software kit

Teamcenter supplies templates to organize sets of definitions for specific industries, applications, or functional areas. Examples of templates provided by Teamcenter are Teamcenter Foundation and Wire Harness Configuration.

These templates are available in the software kit (the installation DVD or downloaded software). For example, the Teamcenter Foundation feature and the Wire Harness Configuration feature are both installed using the software kit, both add definitions to the data model, and both have associated templates found in the software kit.

Templates supplied by asynchronous Teamcenter releases

Some Teamcenter features are released asynchronously and are distributed separately from the Teamcenter software kit (the installation DVD or downloaded software) and are found on their own software kit. An example is the CATIA integration.

Templates supplied by other Teamcenter sites, partners, or third parties

You can use templates created by other sites, partners, or third parties.

Business Modeler IDE

The Business Modeler Integrated Development Environment (IDE) is a tool for customizing the data model of your Teamcenter installation. Use the IDE to create business objects, classes, attributes, lists of values (LOVs), and rules.

The Business Modeler IDE is built on top of the Eclipse platform. Eclipse is a generic platform for tool development that is extended via its plug-in and extension point technology.

You can install the Business Modeler IDE two ways:

- As a stand-alone application
This method installs the Eclipse platform.
- As part of an existing Eclipse environment

Using the Upgrade Assistant

Teamcenter provides a tool, the *Upgrade Assistant* utility, to help you determine impact to your server side customization extension applications when you upgrade to a target release. It evaluates whether custom code needs to be reworked before upgrade. You can evaluate the tool's recommended remediation, for example, ensuring API stability or removing calls to deprecated or obsolete APIs.

This tool currently reports only the usage of deprecated and obsolete ITKs in server side customization applications. The utility takes server side customization application libraries (**dlls**) as input and runs a **dumpbin** report on the library looking for import symbols. The import symbols are then analyzed with the deprecated and obsoleted API in the target release to produce a **csv** file report indicating whether rework is needed immediately (if the application is using an obsolete ITK) or future rework should be planned (if the application is using a deprecated ITK).

The Änderungen in Teamcenter-APIs reference on Support Center identifies replacements for deprecated and obsolete APIs. This reference also provides replacement information for other deprecated and obsolete artifacts like functions and macros. Use this reference when updating your customizations in preparation for upgrade.

Run the Upgrade Assistant as part of your upgrade planning, to help identify and correct customization problems before you begin upgrading your Teamcenter environment. You can run this utility before you download your target upgrade version of Teamcenter.

Download the Upgrade Assistant

Vorgehensweise

1. Log on to Support Center and browse to the Teamcenter 2606 downloads page.
2. Browse to **Additional Downloads**→**Tools for Teamcenter**→**Tools for Teamcenter Upgrade Assistant**.
3. In the resulting list of downloads, find the Upgrade Assistant, and then download the Tc2606_UpgradeAssistantITKReporter.zip package for your platform (Windows or Linux).

Install the Upgrade Assistant (Windows)

Vorgehensweise

1. Make sure a supported version of Microsoft Visual Studio is installed on your local machine. For certified versions of Visual Studio, see the Hardware and Software Certifications knowledge base article on Support Center.
2. Make sure the following system environment variables are set:
MSDEV_HOME

Set to the Visual Studio installation directory, for example, set **MSDEV_HOME=C:\apps\MVS16\VC**.

PATH

Make sure this value includes your Teamcenter Perl install directory, for example, set
PATH=C:\apps\tc\tc2606\TR\perl\bin;%PATH%.

- Expand the Tc2606_UpgradeAssistantITKReporter.zip package to a local directory.
- Open a Teamcenter command prompt from the Windows program list by choosing **Teamcenter**→**Tc-config-name Command Prompt**.¹
- Change to the directory in which you expanded the Upgrade Assistant package:

```
cd <upgrade-assistant-dir>
```

- Type the following command to launch the Upgrade Assistant and view its arguments and options:

```
Tc2606_UpgradeAssistantITKReporter\ITKReporter\bin\TcUpgradeAssistantITKReporter.bat -h
```

Install the Upgrade Assistant (Linux)

Vorgehensweise

- Expand the Tc2606_UpgradeAssistantITKReporter.zip package to a local directory.
- Change to the directory in which you expanded the Upgrade Assistant package:

```
cd <upgrade-assistant-dir>
```

- Run the following command to launch the Upgrade Assistant and view its arguments and options:

```
./ITKReporter/bin/TcUpgradeAssistantITKReporter.sh -h
```

Generate an Upgrade Assistant report

The Upgrade Assistant utility accepts the following arguments:

Argument	Description
apps	Specifies a single path or a sequence of comma-separated file paths, or a text file containing a list of custom applications (absolute path of one

¹ Alternatively, you can set Teamcenter environment variables in a standard command prompt by typing **<TC_DATA>t看_profilevars**, replacing **<TC_DATA>** with the path to your Teamcenter data directory.

Argument	Description
	application per line) or a directory path containing a list of custom applications.
from_release	Specifies your current Teamcenter version, for example, 2606. This argument is optional. If this argument is not provided, the utility reads the Teamcenter version from the environment variables in the Teamcenter command prompt.
out	Specifies the output file (csv format) and path for the Upgrade Assistant report. For example: D:\Temp\mycustom_13_2606.csv
h	Displays help for the utility.

To generate an Upgrade Assistant report, perform the following steps:

Vorgehensweise

1. Open a Teamcenter command prompt from the Windows program list by choosing **Teamcenter** → **Tc-config-name Command Prompt**.
2. Change to the `<upgrade-assistant-dir>\Tc2606_UpgradeAssistant\ITKReporter\ITKReporter\bin` directory (on Windows systems) or `<upgrade-assistant-dir>/Tc2606_UpgradeAssistant\ITKReporter\ITKReporter/bin` (on Linux systems).
3. Type the following command:

- **Windows:**

```
TcUpgradeAssistantITKReporter.bat -apps=<custom-path> -from_release=<current-ver>
-out=<report-path>
```

- **Linux:**

```
TcUpgradeAssistantITKReporter.sh -apps=<custom-path> -from_release=<current-ver>
-out=<report-path>
```

In the appropriate command, replace `<custom-path>` with the path or paths to your custom application **dll** files. Replace `<current-ver>` with your current Teamcenter version. (This argument is optional.²) Replace `<report-path>` with the location in which to generate the report, in **csv** format.

² If **from_release** is not specified, the Teamcenter version is read from the environment variables in the Teamcenter command prompt.

Examples (Windows):

- To generate an Upgrade Assistant report in the current directory:

```
TcUpgradeAssistantITKReporter.bat -apps=D:\\myDir\\MyCustom
```

- To generate an Upgrade Assistant report into a specific directory:

```
TcUpgradeAssistantITKReporter.bat -apps=D:\\myDir\\MyCustom -out=D:\\Temp\\mycustom_13_2606.csv
```

- To generate an Upgrade Assistant report for a Teamcenter version other than the version specified in the Teamcenter command prompt:

```
TcUpgradeAssistantITKReporter.bat -apps=D:\\myDir\\MyCustom -from_release=13.0 -out=D:\\Temp\\mycustom_13_2606.csv
```

Examples (Linux):

- To generate an Upgrade Assistant report in the current directory:

```
TcUpgradeAssistantITKReporter.sh -apps=/home/myusername/myDir/MyCustom.so -from_release=11.0 -out=<output directory>\\mycustom_13_2606.csv
```

- To generate an Upgrade Assistant report into a specific directory:

```
TcUpgradeAssistantITKReporter.sh -apps=/home/myusername/myDir -from_release=11.0 -out=<output directory>\\mycustom_13_2606.csv
```

- To generate an Upgrade Assistant report for a Teamcenter version other than the version specified in the Teamcenter command prompt:

```
TcUpgradeAssistantITKReporter.sh -apps=/home/myusername/myfile.txt -from_release=11.0 -out=<output directory>\\mycustom_13_2606.csv
```

Using the Instance Data Sync utility

The Instance Data Sync utility is a useful tool that can minimize downtime during Teamcenter release updates.

Overview

Many organizations follow the *blue-green* deployment update process to perform updates from one Teamcenter release to another in their production environments. This involves creating a new *green*

Teamcenter environment from a production clone while the *blue* environment remains active. By shifting some deployment tasks outside of the maintenance window if they do not impact production operations, this method can reduce the length of the typical maintenance window..

The *Instance Data Sync* utility is a Teamcenter user data synchronization utility designed to further reduce the maintenance window needed for blue-green deployments. It allows users to remain active in the blue production environment significantly longer and facilitates data replication between the active blue source database and its cloned green target database or identical types. The utility leverages database change tracking triggers and recipe-driven data transformation mechanisms to synchronize database changes to the upgraded green environment.

You can apply these benefits to a Teamcenter **Platform Update** or **Basic Update** by incorporating the Instance Data Sync utility into your upgrade process.

Key features

- Change tracking: Monitors data modifications in Teamcenter database using lightweight triggers
- Recipe-based transformation: JSON-configurable data morphing between schema versions
- Cross-database support: Oracle and Microsoft SQL Server compatibility
- Version migration: Handles schema evolution through recipe positioning

Managing the database and volume data during blue-green deployment

Blue-green deployment requires creating a clone of the database from the source environment. The Instance Data Sync utility processes Teamcenter database content, but does *not* process Teamcenter volume files. Volumes can be managed by sharing user volumes between Blue and Green environments and using a copy of the Blue environment's **dba** volume for the Green environment's **dba** volume. However, it should be noted that if you use a shared volume, you should not modify production volume data in the Green environment while you are testing it. Alternatively, a complete copy of the Blue environment's volumes can be used as the volumes for the Green environment.

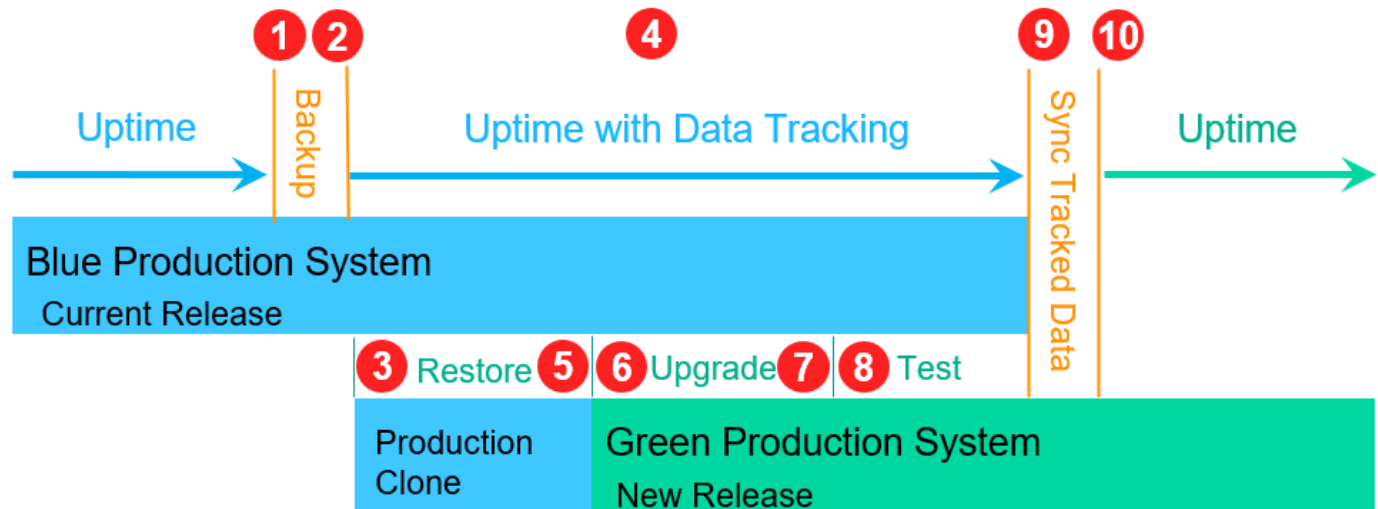
When you create the Green environment, first restore the database clone from the Blue environment that will be used by the Green environment. Next, depending on your volume approach, ensure your copied Blue **dba** volume or full volumes are available for deploying the Green environment.

Take care during testing of the updated Green environment. To avoid item ID conflicts, restrict item creation, use a different naming range, or restore the clone and sync user data again after testing is complete. If you use a shared volume, do not modify production volume data in the Green environment while you are testing it.

Before making the Green environment active, synchronize volumes by copying the volumes from the production environment in the intermediate environment machine. This is not required if you use a shared volume.

Platform update process with Instance Data Sync

Instance Data Sync facilitates the platform update process by ensuring data consistency and minimizing downtime during Teamcenter upgrades.



A blue-green update using the Instance Data Sync utility may proceed as follows:

- 1 In the source environment, or *Blue* environment, back up the source database to create a cloned database. Share or copy volumes from the Blue environment. Users must log off before this action.
- 2 Run the Instance Data Sync **StartSource** action on the source, or Blue, database. Users may log on and resume working in the Blue environment. The Blue environment remains active and users can continue to create data at this time, because data changes are tracked.
- 3 Create an intermediate environment, or *Green* environment: Restore the database clone from the Blue environment, make the Blue environment's volumes available to the Green environment, and then build and deploy the Green environment using either the Production Clone option in Deployment Center or the **platform update** process.
- 4 Users continue working the Blue environment while the update process proceeds in the Green environment.
- 5 Register a TEM-deployed environment or upgrade Deployment Center for a Deployment Center-deployed environment.
- 6 Run the Instance Data Sync **StartDestination** action on the intermediate (Green) database.
- 7 Perform the Teamcenter update to the target version. Back up the upgraded database.
- 8 Smoke test the updated environment to verify that the update was successful.
- 9 Run the Instance Data Sync **Sync** action on the intermediate (Green) database to synchronize data changes from the Blue environment to the Green environment. Users must log off before this action.
- 1 Make the Green environment active so users may log on and resume work. The Blue environment can then be retired.
- 0

Instance Data Sync actions

The utility performs four distinct actions that comprise a complete synchronization workflow.

StartSource

The *StartSource* action prepares the source database for change tracking. It creates the **INSTANCEDATASYNCCHANGES** table to track modifications. It installs triggers on all lightweight POM classes and **POM_object**. It begins monitoring **INSERT**, **UPDATE**, and **DELETE** operations.

Usage in Oracle:

```
InstanceDataSync.exe --action StartSource --sourceDbType "Oracle"
--sourceDbServer "<source-database-host>" --sourceDbUsername "<source-
database-user>" --sourceDbPassword "<source-database-user-password>" --
sourceDatabase "<source-database-schema>" --sourceSsid "<source-database-
SID>" --log "<logs-directory>"
```

Usage in Microsoft SQL Server:

```
InstanceDataSync.exe --action StartSource --sourceDbType "SqlServer"
--sourceDbServer "<source-database-host>" --sourceDbUsername "<source-
database-user>" --sourceDbPassword "<source-database-user-password>" --
sourceDatabase "<source-database-schema>" --log "<logs-directory>"
```

StartDestination

The *StartDestination* action prepares the destination database after restoration. It removes any existing sync triggers from the restored database. It cleans up the sync changes table, if present. It advances the UID generator to prevent conflicts with future source changes.

Usage in Oracle:

```
InstanceDataSync.exe --action StartDestination --sourceDbType "Oracle"
--sourceDbServer "<source-database-host>" --sourceDbUsername "<source-
database-user>" --sourceDbPassword "<source-database-user-password>" --
sourceDatabase "<source-database-schema>" --sourceSsid "<source-database-
SID>" --log "<Logs directory>"
```

Usage in Microsoft SQL Server:

```
InstanceDataSync.exe --action StartDestination --sourceDbType "SqlServer"
--sourceDbServer "<source-database-host>" --sourceDbUsername "<source-
database-user>" --sourceDbPassword "<source-database-user-password>" --
sourceDatabase "<source-database-schema>" --log "<Logs directory>"
```

Sync

The *Sync* action synchronizes accumulated changes from the source to the destination. It reads change records from the source sync table and groups changes by class and operation type. It applies recipe-based transformations, if configured. It replicates changes to the destination database. It processes **INSERT**, **UPDATE**, and **DELETE** operations in the correct order.

Usage in Oracle:

```
InstanceDataSync --action Sync --sourceDbType Oracle --sourceDbServer
<source-server> --sourceDbUsername <source-username> --sourceDbPassword
<source-password> --sourceSsid <source-sid> --destinationDbType Oracle --
destinationDbServer <dest-server> --destinationDbUsername <dest-username> --
destinationDbPassword <dest-password> --destinationSsid <dest-sid>
```

Usage in Microsoft SQL Server:

```
InstanceDataSync --action Sync --sourceDbType SqlServer --sourceDbServer
<source-server> --sourceDbUsername
<source-username> --sourceDbPassword <source-password> --sourceDatabase
<source-database> --destinationDbType
SqlServer --destinationDbServer <dest-server> --destinationDbUsername
<dest-username> --destinationDbPassword
<dest-password> --destinationDatabase <dest-database>
```

FinishSource

The *FinishSource* action removes synchronization infrastructure from the source database. This is an optional step. This action drops all sync triggers from the source database. It removes the **INSTANCEDATASYNCCHANGES** table, and it cleans up any temporary synchronization objects.

Usage in Oracle:

```
InstanceDataSync.exe --action FinishSource --sourceDbType "Oracle"
--sourceDbServer "<source-database-host>" --sourceDbUsername "<source-
database-user>" --sourceDbPassword "<source-database-user-password>" --
sourceDatabase "<source-database-schema>" --sourceSsid "<source-database-
```

```
SID>" --log "<Logs directory>"
```

Usage in Microsoft SQL Server:

```
InstanceDataSync.exe --action FinishSource --sourceDbType "SqlServer"
--sourceDbServer "<source-database-host>" --sourceDbUsername "<source-
database-user>" --sourceDbPassword "<source-database-user-password>" --
sourceDatabase "<source-database-schema>" --log "<Logs directory>"
```

Recipe configuration

Recipes enable data transformation during synchronization, allowing schema evolution and data morphing between different Teamcenter versions.

Recipe structure

Recipes are JSON files that need to be provided as part of the `--recipe` argument to the sync process, which has the files with the naming pattern `*_maprecipe.json`:

```
[
  {
    "version": "<Tc-version-when-schema-changed>",
    "pomClass": "<POM-class-of-new-schema>",
    "elements": [
      {
        "type": "<types-supported-by-IDS-utility>",
        "attributeName": "<attribute-name-on-which-to-populate-data>",
        "expression": "C# LINQ expressions with x object context - Built-in
functions to string manipulation"
      }
    ]
  }
]
```

Example recipe

Here is a practical example showing object type string manipulation:

```
[
  {
    "version": "13.1",
    "pomClass": "Cst0Base",
```

```

    "elements": [
      {
        "type": "SetAttribute",
        "attributeName": "cst0RevisionAnchor",
        "expression": "x.cst0IRDI.Replace(\"#01-\", \"#\") "
      }
    ]
  }
]

```

This recipe transforms the **cst0RevisionAnchor** attribute by populating values from the **cst0IRDI** values. It does this by replacing **#01-** prefixes with **#** for all the **Cst0Base** instances during the version 13.1 synchronization.

Recipe elements

SetAttribute

Sets attribute values using dynamic expressions:

```

{
  "type": "SetAttribute",
  "attributeName": "propertyName",
  "expression": "x.sourceProperty + \" modified\""
}

```

VlaToRelation / RelationToVla

Converts between variable-length arrays and relation objects:

```

{
  "type": "VlaToRelation",
  "name": "conversionName",
  "attributeName": "vlaAttribute",
  "relationClass": "TargetRelationClass"
}

```

Recipe filtering

The morphing engine supports sophisticated filtering capabilities that can be combined to target specific data subsets while preserving all other rows during synchronization.

Global filter

Applied to all rows using dynamic LINQ expressions:

```
{  
  "filter": "x.status == 'Active' && x.revision > 0"  
}
```

Filter processing behavior

- Matching Rows: Processed through recipe transformations
- Non-Matching Rows: Preserved unchanged and merged back into results
- Zero Data Loss: All rows are synchronized regardless of filter criteria

Expression syntax

Recipe expressions use dynamic LINQ syntax with access to source object properties. For comprehensive string manipulation functions and expression syntax, refer to the **System.String** methods documentation from Microsoft.

Recipe versioning and processing

Recipes define how data should be transformed when synchronizing between source and destination environments running different Teamcenter releases.

Instance Data Sync supports synchronization between the following Teamcenter versions:

- 13.x Series: 13.0, 13.1, 13.2, 13.3
- 14.x Series: 14.0, 14.1, 14.2, 14.3
- 23.x Series: 2306, 2312
- 24.x Series: 2406, 2412
- 25.x Series: 2506, 2512
- 26.x Series: 2606

When synchronizing between versions, for example, from Teamcenter 13.0 to 2506, the system:

- Identifies all intermediate versions between the source and the target.
- Applies transformation recipes in chronological order.

Command line parameters

Required parameters

Parameter	Description
--action	Action to perform: StartSource , StartDestination , Sync , and FinishSource
--sourceDbType	Source database type: Oracle or SqlServer

Source database parameters

Parameter	Oracle	Microsoft SQL Server	Description
--sourceDbServer	✓	✓	Database server hostname or IP
--sourceDbUsername	✓	✓	Database username
--sourceDbPassword	✓	✓	Database password
--sourceSsid	✓	✗	Oracle SID or Service Name
--sourceDatabase	✗	✓	Microsoft SQL Server database name
--sourcePort	✓	✓	Database port (default: 1521 for Oracle)

Destination database parameters (Sync action only)

Parameter	Oracle	Microsoft SQL Server	Description
--destinationDbType	✓	✓	Destination database type
--destinationDbServer	✓	✓	Destination server hostname or IP
--destinationDbUsername	✓	✓	Destination username
--destinationDbPassword	✓	✓	Destination password
--destinationSsid	✓	✗	Oracle destination SID
--destinationDatabase	✗	✓	Microsoft SQL Server destination database
--destinationPort	✓	✓	Destination port

Other parameters

Parameter	Description
--recipe	Directory containing recipe JSON files. This is for sync action only
--log	Directory for log file output. The default is the current directory.

Troubleshooting

Common issues

- Connection Failures: Verify database connectivity, credentials, and network access.
- Permission Errors: Ensure database user has DDL privileges for trigger creation.
- Recipe Errors: Validate JSON syntax and expression compatibility.
- Sync Conflicts: Check for concurrent modifications and UID conflicts.

Best practices

- Both Blue and Green databases should point to the same volumes.
- Post-upgrade manual processing steps: Perform the required post-upgrade processing steps after running the Instance Data Sync utility. This includes running any manual utilities explicitly mentioned in the release notes. For example, run the **add_update_fnd0impactingobjects** utility for the Configurator module.
- Do not perform a BMIDE deployment to the blue environment during this process.
- Backup before start: Always back up the source database before running StartSource, and backup the destination database before running Sync.
- Monitor logs: Review the log output for errors.
- Help message: Run the utility with **--help** to display a help message.

Install the Siemens License Server

Before you update Teamcenter, you must download and install the [Siemens License Server](#) to distribute licenses to Teamcenter hosts. Even if you already use the Siemens License Server with your current version of Teamcenter, you must install the version of the License Server supported by the version of Teamcenter to which you update.

For the version of the Siemens License Server certified with Teamcenter 2606, see the Hardware and Software Certifications knowledge base article on Support Center.

Download and install the Siemens License Server:

Vorgehensweise

1. Open Support Center:
<https://support.sw.siemens.com>
2. Under **Product Centers**, find **Siemens License Server**.

Achtung

Make sure you download **Siemens License Server**, *not* **Siemens PLM Licensing**.¹

3. In the Siemens License Server product center, click **Downloads**, and then download the certified version of the Siemens License Server.
4. Install the License Server according to the *Siemens Digital Industries Software License Server Installation Instructions* available from the Siemens License Server downloads page.
5. On your designated Teamcenter corporate server host, set the following system environment variables:

SPLM_LICENSE_SERVER

Set to the location of the Siemens License Server:

```
<port>@<host>
```

Replace *<port>* with the port number and *<host>* with the machine name of the License Server, for example, **29000@tchost**.

The value of this variable is designated as the default local license server during corporate server upgrade.

TCP_NODELAY

Set to a value of **1** on the License Server host. This helps optimize logon time when launching Teamcenter.

6. Install Teamcenter licenses on the License Server according to the information provided to you by Siemens Digital Industries Software support.

Nächste Maßnahme

The **Siemens License Server** downloads page contains additional links to documentation, Knowledge Base articles, and videos about installing and maintaining the License Server.

¹ Siemens PLM Licensing is no longer supported by Teamcenter. The Siemens License Server is the currently supported license server.

Upgrading database servers

Supported database server versions

If your database server version is not a version certified for Teamcenter 2606, you must upgrade your database server before you upgrade Teamcenter.

Siemens Digital Industries Software supports Teamcenter 2606 with Oracle and Microsoft SQL Server databases. For supported versions of Oracle and Microsoft SQL Server, see the Hardware and Software Certifications knowledge base article on Support Center.

Configuring character settings in non-English locales

To ensure correct display and processing of Teamcenter data, you must set required values in your system environment and your Teamcenter configuration.

If your Teamcenter servers run Linux and use the Unicode UTF-8 character set, you must configure your operating system and your Teamcenter configuration to use Unicode UTF-8, observing the following considerations during Teamcenter upgrade:

- An existing Teamcenter installation must not configure Unicode UTF-8 character set support with an upgrade to Teamcenter 2606. An existing Teamcenter installation is required to complete an upgrade to Teamcenter 2606 under its existing database character set encoding.
- After upgrade is completed, you must contact your database vendor for processes and tools to convert your existing database character set encoded data to Unicode UTF-8 character set encoded data for storage to a new database.

For more information about environment settings for your locale, platform, and character set, see UTF-8 settings for Windows or Linux systems, or non-UTF-8 settings for Windows or Linux systems in the Teamcenter installation guides.

Set shell limits and parameters for Oracle on Linux

Overview of shell limits and parameters

Oracle RDBMS uses extensive Linux resources such as shared memory, swap memory, and semaphore for interprocess communication. Inadequate parameter settings cause problems during installation and startup. Increasing the volume of data stored in memory reduces disk I/O activity and improves database performance.

The Oracle RDBMS installation program displays warnings if kernel parameters are not adequate. To avoid warnings and errors during or after installation, make sure kernel parameters meet the recommended settings for typical environments described in the following topics.

Before you install Oracle RDBMS, set initial parameters as described in Oracle documentation, and then adjust parameters according to available system memory. Set the **ulimit** parameter to **unlimited**.¹ Then, set the **kernel parameters** to recommended Teamcenter values for your operating system.

If you previously tuned kernel parameters for other installed applications to levels that meet or exceed the values recommended for Teamcenter, keep those existing values.

The parameter settings recommended herein are *minimum* values. For production database systems, Oracle recommends you tune values to optimize system performance. For information about performance tuning, see:

- Documentation for your operating system
- Teamcenter installation documentation on Support Center

Set SUSE Linux shell limits

Vorgehensweise

1. Increase shell limits for the **oracle** user to the minimum values listed in the following table by adding the following lines to the **/etc/security/limits.conf** file:

oracle	soft	nproc	2047
oracle	hard	nproc	16384
oracle	soft	nofile	1024
oracle	hard	nofile	65536

Do not change the shell limit values if they were set for another program and the values are greater than the levels Oracle requires.

SUSE Linux shell limit	Item in limits.conf	Minimum hard limit
Maximum number of open file descriptors	nofile	65536
Maximum number of processes available to a single user	nproc	16384

2. Add or edit the following lines in the **/etc/pam.d/login** file:

```
session required /lib64/security/pam_limits.so
session required pam_limits.so
```

3. Change the **oracle** user default shell startup file:

¹ The **ulimit** parameter specifies a maximum number of processes per user.

- For the Bourne, Bash, or Korn shell, add the following lines to the **/etc/profile.local** file:

```
if [ $USER = "oracle" ]; then
    if [ $SHELL = "/bin/ksh" ]; then
        ulimit -u 16384
        ulimit -n 65536
    else
        ulimit -u 16384 -n 65536
    fi
fi
```

- For the C shell (csh or tcsh), add the following lines to the **/etc/csh.login.local** file:

```
if ( $USER == "oracle" ) then
    limit maxproc 16384
    limit descriptors 65536
endif
```

Upgrade an Oracle server and database

Exporting an Oracle database

Export an Oracle database (Windows systems)

Vorgehensweise

1. Log on to the Oracle server as an administrator user.
2. Export the contents of your Teamcenter Oracle database to the dump file:

```
<ORACLE_HOME>\bin\expdp <db-user>/<password> full=y dumpfile=<file-name>.dmp logfile=<export>.log
```

Replace **<db-user>** with the Teamcenter database user account name; replace **<password>** with the database user account password; replace **<file-name>** with the full path and name of the dump file to contain the exported data; replace **<export>** with the name of the log file to contain export output.

3. Store the dump file in a safe place.

Nächste Maßnahme

If you have multiple databases, repeat this procedure for each database.

Achtung

Siemens Digital Industries Software strongly recommends backing up the dump file on tape or another disk. If the dump file becomes corrupted or lost, all data from the existing database is lost.

Export an Oracle database (Linux systems)

Vorgehensweise

1. Either log on to the Oracle server as **oracle** or switch the user to **oracle**:

```
su - oracle
```

2. Set the **PATH** environment variable to include the Oracle **bin** directory:

```
export PATH=$PATH:<ORACLE_HOME>/bin
```

3. Manually set the shared library path for Linux:

```
export LD_LIBRARY_PATH=${LD_LIBRARY_PATH}:${<ORACLE_HOME>}/lib
```

4. Export the contents of the Teamcenter Oracle database to the dump file:

```
<ORACLE_HOME>/bin/exp <db-user>/<password> full=y file=<file-name>.dmp  
log=<export>.log
```

Replace *<db-user>* with the Teamcenter database user account name; replace *<password>* with the database user account password; replace *<file-name>* with the name of the dump file to contain the exported data; replace *<export>* with the name of the log file to contain export output.

5. Store the dump file in a safe place.

Nächste Maßnahme

If you have multiple databases, repeat this procedure for each database.

Achtung

Siemens Digital Industries Software strongly recommends backing up the dump file on tape or another disk. If the dump file becomes corrupted or lost, all data from the existing database is lost.

Terminate Oracle sessions on Windows systems

Stop the listener process

Vorgehensweise

1. Log on to the operating system as a user with administrator privileges.
2. Open the **Services** dialog box in the Windows Control Panel.
3. Select the Oracle TNS listener services (**Oracle<release-ID>TNSListener**) and click **Stop**.

Shut down an Oracle database

Shut down Oracle using Windows Control Panel

Vorgehensweise

1. Log on to the operating system as a user with administrator privileges.
2. Open the **Services** dialog box in the Windows Control Panel.
Windows displays the Services window.
3. Select the **OracleService<SID>** service.
Replace <SID> with the system identifier of the database instance.
4. Click **Stop**.

Shut down Oracle using SQL*Plus

Vorgehensweise

1. Log on to the operating system as a user with administrator privileges.
2. Start the Oracle SQL*Plus utility:

```
sqlplus sys/<password>@<Oracle-SID> as sysdba
```

Replace <password> with the password for the **sys** user account.

Oracle starts the Oracle SQL*Plus utility.

The **sys** user must be in the Oracle **sysdba** group for the Oracle system identifier (SID) used by Teamcenter. To connect as internal (without a password), the account must be part of the **ORA_DBA** local group in Windows.

3. Shut down the database instance by typing the following command:

```
shutdown
```

4. Exit SQL*Plus:

```
exit
```

Terminate Oracle sessions on Linux systems

Before installing a new version of Oracle, you must terminate all Oracle sessions and Oracle processes.

Vorgehensweise

1. Either log on to the Oracle server as **oracle** or switch the user to **oracle** as follows:

```
su - oracle
```

2. Set the **ORACLE_HOME** environment variable to point to the location of the Oracle files. For example:

```
export ORACLE_HOME=/u01/app/oracle/product/<oracle-version>
```

Replace the path with the system path to the Oracle files.

3. Define **ORACLE_HOME/bin** in the **PATH** variable:

```
export PATH=${PATH}:${ORACLE_HOME}/bin
```

4. Manually set the shared library path on Linux:

```
export LD_LIBRARY_PATH=${LD_LIBRARY_PATH}:${ORACLE_HOME}/lib
```

5. If a **tnslsnr** listener process is running, terminate it. For example:

```
$ORACLE_HOME/bin/lsnrctl stop <listener-name>
```

Replace *<listener-name>* with the name of the listener process.

6. Shut down all Oracle database instances using the **dbshut** utility. Shut down database instances listed in the **oratab** file:

```
$ORACLE_HOME/bin/dbshut
```

Back up an Oracle installation

If you are upgrading to the certified Oracle version, back up the existing Oracle installation.

Backing up your Oracle installation before upgrading is strongly recommended. Failure to back up existing data could result in loss of data if problems occur during the upgrade process.

Back up the following files and directories:

- The Oracle home directory on each installed workstation.
- The directories containing database files for each configured database.
- The Oracle Net **listener.ora** and **tnsnames.ora** configuration files in the **/etc** directory.

These are the only Teamcenter directories affected by Oracle installation. If you created other directories containing data used by Oracle, such as an administration script directory, you should also back up these directories.

Upgrading an Oracle server

Upgrade the Oracle server

Upgrade your Oracle server by one of the following methods:

- *Upgrade using the Oracle installer*
- *Upgrade by uninstalling/reinstalling Oracle*

Upgrade using the Oracle installer

Vorgehensweise

1. Launch the Oracle installer to install a certified version of Oracle server.
2. When the Oracle installer prompts you to upgrade existing databases, enter the required information about the databases you want to upgrade.

Installing an Oracle server is described in the Teamcenter installation guides for Windows and Linux.

Upgrade by uninstalling/reinstalling Oracle

Vorgehensweise

1. Remove existing Oracle databases.
2. Uninstall all existing Oracle server software.

3. Install a certified version of Oracle server.

Installation of an Oracle server is described in the Teamcenter installation guides for Windows or Linux.

4. After Oracle installation is complete, import your Teamcenter database from the Oracle dump file into the new Oracle database. Enter the following command on a single line:

```
<ORACLE_HOME>\bin\imp <db-user/password> fromuser=<db-user> touser=<db-user> file=<file-name>.dmp log=<import>.log
```

Replace *<db-user>* with the Teamcenter database user account name, *<password>* with the database user account password, *<file-name>* with the full path and name of the dump file that contains the exported data, and *<import>* with the name of the log file.

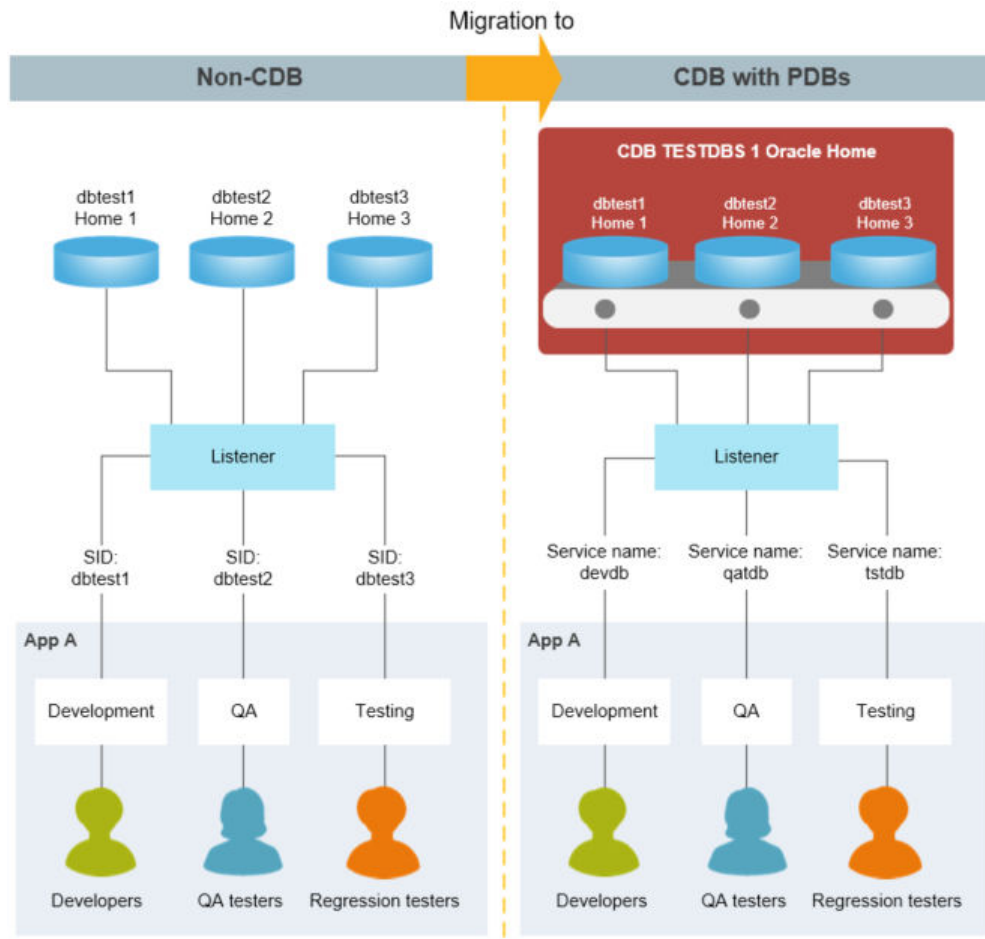
Migrate a non-CDB database to a CDB database

Teamcenter supports Oracle's [multitenant database architecture](#) if you use Oracle 12c or later. A multitenant architecture is deployed as a Container Database (CDB) with one or more Pluggable Databases (PDB).

A *Container Database* (CDB) is similar to a conventional (non-CDB) Oracle database, with familiar concepts like control files, data files, undo, temp files, redo logs, and so on. It also houses the data dictionary for objects owned by the root container and those that are visible to databases in the container.

A *Pluggable Database* (PDB) contains information specific to the database itself, relying on the container database for its control files, redo logs and so on. The PDB contains data files and temp files for its own objects, plus its own data dictionary that contains information about objects specific to the PDB. From Oracle 12.2 onward a PDB can and should have a local undo tablespace.

You can [migrate a non-CDB database to a CDB database](#) using Oracle tools. The following example illustrates the database architectures before and after migration.



Teamcenter supports CDB and non-CDB databases. Be aware that **Oracle has deprecated support for non-CDB databases** and may discontinue support after Oracle 19c.

If you migrate a non-CDB Teamcenter database to a CDB database, you must perform the migration *after* you upgrade to Teamcenter 2606.

Upgrade Microsoft SQL Server databases

Upgrading a Microsoft SQL Server database server with an existing Teamcenter database requires exporting the contents of the existing database, installing the new Microsoft SQL Server database server, and importing the database into the new server.

Microsoft describes several methods for transferring databases between servers in the article titled *How to move databases between computers that are running SQL Server* at <https://support.microsoft.com>.

Siemens Digital Industries Software recommends a backup and restore method for transferring an existing Teamcenter database to a new Microsoft SQL Server server. One advantage of this method is the simplicity of a single backup file, rather than multiple files (.mdf, .ndf, and so on), as with an attach/detach method.

A single backup file also minimizes potential compatibility mode issues between versions of Microsoft SQL Server.

Using Data Transformation Services (DTS) for transferring a database is the least recommended method because is relatively more time-consuming and requires communication between the old and new database servers. In contrast, other methods for database transfer require only a means to move export files from one server to the other.

Siemens Digital Industries Software recommends creating the Teamcenter database user in the new target database *before* importing the existing database into the new server.

For transferring logins and passwords to a new Microsoft SQL Server instance, Microsoft provides an article titled *How to transfer logins and passwords between instances of SQL Server* at <https://support.microsoft.com>.

To ensure the owner of the database is the new server's Teamcenter database user account rather than the old server's Teamcenter database user account,¹ enter the following commands:

```
exec sp_changedbowner 'sa'  
exec sp_changedbowner '<Tc-db-user>'
```

¹ Internal integer IDs used by SQL Server can differ between sites.

8. Build a target pre-production environment

Create a Teamcenter 2606 pre-production environment that contains the same applications, components, and custom software as your production environment.

Prepare the production environment for export

Terminate Teamcenter sessions

Prior to upgrade, you must terminate Teamcenter sessions if:

- You are reinstalling or upgrading Teamcenter executables by overwriting an existing Teamcenter data directory. The Teamcenter installation procedure cannot overwrite files when they are in use.
- You are upgrading a Teamcenter database.
- You are migrating an Oracle database to a Windows database server.

Vorgehensweise

1. Instruct all users to check in all Teamcenter business objects, and then close and log off of Teamcenter sessions, including **tcserver** processes.
2. Open a Teamcenter command prompt.
3. Use the **clearlocks** utility to check for nodes connected to the database and remove locks on the database:

Windows systems:

```
%TC_BIN%\clearlocks -u=<Tc-Oracle-user> -p=<Tc-Oracle-user-password>
-g=dba -assert_all_dead
```

Linux systems:

```
$TC_ROOT/bin/clearlocks -node_names
```

4. On Linux systems, note the node names returned, and then type the following command for each node name returned:

```
$TC_ROOT/bin/clearlocks -assert_dead <node-name>
```

Replace `<node-name>` with a returned node name.

5. Stop all Teamcenter services, including FMS.

Preserve content security policy directives

If you created content security policy (CSP) directives in a Teamcenter 2512 environment, you must capture those directives and re-enter them in the updated Teamcenter environment after update.

Procedure

1. Open the `microservices/gateway/config.json` file in your current Teamcenter environment.
2. Record all CSP directive IDs and their corresponding URLs. For example:

```
"security" : {
  ...
  "cspPolicy": {
    "directives" : {
      ...,
      ...,
      "frame-src": [
        "https://*.siemens.com", // add the URL that
needs to be whitelisted. Note, this should be done keeping application
security in mind
      ],
    }
  }
  ...
}
```

What to do next

After you complete the update of your Teamcenter environment, re-enter your CSP directives in the **Active Workspace Gateway** component as described in Content security policy directive configuration in *Teamcenter-Sicherheit*.

Back up existing Teamcenter data

Before you upgrade a Teamcenter database, back up existing Teamcenter data.

Achtung

Back up the database, Teamcenter data directory, and all Teamcenter volume directories to an external backup device before performing an upgrade. This provides a safeguard against data loss in case problems occur during the upgrade.

Back up the following directories:

- The Teamcenter application root directory on each installed workstation
- The Teamcenter data directory for each configured database
- The Teamcenter volume directories for each configured database

These are the only directories affected by Teamcenter installation. If you created other directories that contain data used by your existing Teamcenter installation, such as a separate POM transmit schema directory, Siemens Digital Industries Software recommends that you back up these directories as a precautionary measure.

Clean unused columns from the database

If you use an Oracle database, upgrade performance may decrease when dropping columns from a large Teamcenter class. This can affect overall upgrade time.

To minimize overall upgrade time, you can mark a column to be dropped as unused by typing the following SQL statement:

```
ALTER TABLE <table-name> SET UNUSED <column-name>
```

This statement marks the column as unused and hides it from any SQL used on that table. Commands like **DESCRIBE** <table_name> or **SELECT * from** <table-name> will not show the column. Any column that is marked as unused is not displayed in queries or data dictionary views and its name is removed so that a new column can reuse the same name. All constraints, indexes, and statistics defined on the column are also removed.

To physically clean up unused columns and reclaim space, use the **install** utility with the **clean_unused_columns** argument:

```
install -clean_unused_columns <Tc-Oracle-user> <password> dba
```

Depending on the number of unused columns in the database and size of their related tables, this command may generate large **redo** logs, especially if large tables are involved. Siemens Digital Industries Software recommends you adjust the size of the **redo** logs appropriately before attempting to use this utility, and make sure you have exclusive access to the schema, for example, during the maintenance window.

Export the production environment

Set default FMS keys

Before you create a copy of the Teamcenter database, set FMS keys to their default values.

Vorgehensweise

If your current production environment is Teamcenter 14.3 or later, before you export Teamcenter Database, set the FMS keys to default. After you export the Teamcenter Database, the keys will be reconfigured to their original values.

To set the FMS keys to default, perform the following steps:

- a. Export the existing non-default FMS keys using one of the following commands, depending on whether symmetric or asymmetric keys are configured:

Symmetric Keys:

```
install_encryptionkeys.exe -u=<Tc-admin> -p=<Tc-admin-pw> -f=list
```

Replace *<Tc-admin>* with the Teamcenter administrative user, and *<Tc-admin-pw>* with the password for that account.

Anmerkung

Make sure you keep a copy of the exported key, similar to the following example (in **bold**):

```
Key_Bits ---- Rijndael Key
128 ---- CFBD7EAA2A90377651E217EF5CC248C7
```

Asymmetric Keys:

```
install_fms_keys.exe -u=<Tc-admin> -p=<Tc-admin-pw> -g=dba
-f=exp_pubkey -file_password=<export-file-password>
```

Replace *<Tc-admin>* with the Teamcenter administrative user, and *<Tc-admin-pw>* with the password for that account.

- b. Delete the existing key configuration:

- **Symmetric Keys:**

```
install_encryptionkeys.exe -u=<Tc-admin> -p=<Tc-admin-pw> -f=delete
```

- **Asymmetric Keys:**

```
install_fms_keys.exe -u=<Tc-admin> -p=<Tc-admin-pw> -g=dba -delete
```

- c. Install the default FMS keys:

```
install_encryptionkeys.exe -u=<Tc-admin> -p=<Tc-admin-pw> -f=install
```

Copy the production database

Copy the production database to back up Teamcenter metadata.

Vorgehensweise

1. Log on to the Oracle server.

Linux systems:

- a. Log on to the Oracle server as **oracle** or switch the user to **oracle**:

```
su - oracle
```

- b. Set the **PATH** environment variable to include the Oracle **bin** directory:

```
export PATH=$PATH:<ORACLE_HOME>/bin
```

- c. Manually set the shared library path for Linux:

```
export LD_LIBRARY_PATH=${LD_LIBRARY_PATH}:${<ORACLE_HOME>}/lib
```

Windows systems:

- a. Log on to the Oracle server as an administrator user.
2. Export the contents of your Teamcenter Oracle database to the dump file:

```
<ORACLE_HOME>\bin\expdp <db-user> /<password> full=y dumpfile=<file-name>.dmp logfile=<export>.log
```

Replace **<db-user>** with the Teamcenter database user account name; replace **<password>** with the database user account password; replace **<file-name>** with the full path and name of the dump file to contain the exported data; replace **<export>** with the name of the log file to contain export output.

3. Store the dump file in a secure place.

Restore FMS keys

Restore your initial FMS keys.

Vorgehensweise

1. Reset the FMS keys back to their original configuration:

- **Symmetric Keys:**

```
install_encryptionkeys.exe -u=<Tc-admin> -p=<Tc-admin-pw> -f=modify
-key=<FMS-key>
```

- **Asymmetric Keys:**

```
install_fms_keys.exe -u=<Tc-admin> -p=<Tc-admin-pw> -g=dba -f=import
-file_name=<PKCS-file>
```

Replace <PKCS-file> with the file path and name of the exported PKCS12 file containing the key pair.

2. Zip the **volume(s)** directory from the original environment.
3. Zip the <TC_DATA>/**lgs_info** directory from the original environment.
4. In the new environment, create the same database user from the original environment.
5. Import the Teamcenter database to the database in the new environment.
6. Unzip the **volume(s)** directory and note the location.
7. Unzip the <TC_DATA>/**lgs_info** directory into the new <TC_DATA> directory on the machine where the environment will be hosted.

Copy Teamcenter volume data

Vorgehensweise

1. Create a copy of your default volume.
2. Create a copy of an additional representative Teamcenter volume for testing.

Export the existing production environment

Vorgehensweise

1. Export the production environment to an XML configuration file using the Deployment Center interface or the quick deploy utility.

- **Using the Deployment Center interface**

On the **Environments** page, choose the environment from the list, and click **Export Environment**

 Select the **Export all component properties**  check box. Save the exported file

- Using the quick deploy utility

- Use the quick deploy utility (**dc_quick_deploy**) in export mode. Type the following command:

```
dc_quick_deploy.bat|sh -dcusername=<DC-user> -dcpassword=<DC-password>  
-mode=export -exportType=Full -environment=<env-name> -dcurl=<DC-URL>
```

2. Open the exported XML configuration file and edit values:

- Update all instances of **REPLACEME** with the correct passwords.
- Edit the machine names in the exported XML configuration file to match the machines that will host the new Teamcenter binaries in the test environment. If you skip this step, you will need to update the machine names in the Deployment Center UI after the test environment is created.

Update any connections, accounts, or UNC paths to reference the target machines as well.

Copy custom extensions

Vorgehensweise

Copy your custom extension files from source control or from the Teamcenter environment.

10. Build a pre-production environment

Import the copy of your Teamcenter production into a pre-production environment in which to develop the update process you will apply to your production environment.

Import the copied environment into the pre-production environment

Install a preproduction Deployment Center server

Vorgehensweise

1. Install a new instance of Deployment Center of the same version as your existing Deployment Center instance that hosts your production environment. Apply any migration hotfixes to the Deployment Center instance.
2. If your environment contains multiple servers, determine whether you want your test environment to include all servers from the original environment or a subset of the servers. If you want to test a subset of the original environment, modify the configuration XML file to remove components you want to omit.

This approach can be helpful when you want to test the production migration in stages, validating each stage, and then adding more components until you import the entire environment.

For a complex environment, if you only need to validate data model related changes, you can remove other components from the configuration XML file, for example, multiple FSCs.

Import the copy of the production database

Vorgehensweise

1. Verify that the time zone setting in your pre-production Oracle server matches that of your production server.

If the time zone of the pre-production database does not match that of the copied database, the import may fail

2. On the pre-production Oracle server, change to the `<ORACLE_HOME>\bin` directory.

3. Enter the following command to import the copy of your production database into the database server for your pre-production environment:

```
impdp <db-user>/<password> SCHEMAS=<db-user> REMAP_SCHEMA=<db-user>:<db-user> DIRECTORY=<file-path> DUMPFILE=<file-name>.dmp  
EXCLUDE=constraint, ref_constraint, index TABLE_EXISTS_ACTION=replace  
LOGFILE=<import-log>.log
```

Replace values in this command as follows:

- <db-user>: Teamcenter database user account name
- <password>: Teamcenter database user password
- <file-path>: Path to the dump file.
- <file-name>: Dump file that contains the exported data.
- <import-log>: Name for the command output log file.

Import the copied environment into Deployment Center

Vorgehensweise

1. Import the configuration XML file into Deployment Center using the quick deploy utility:

```
dc_quick_deploy.bat|sh -dcusername=<DC-user> -dcpassword=<DC-password>  
-inputFile=<config-XML-file> -dcurl=<DC-URL>
```

2. Set the database creation parameters.
 - a. Under **Database Creation Settings**, make sure the **Copy Environment using existing populated database** option is selected.

WARNUNG

When you enter the database and volume information, use the copied Teamcenter database and volume, *not* the original production database and volume.

3. Add the volume information in the respective fields, adding the unzipped **volume(s)** path you entered when you **restored FMS keys**.

The **COPIED VOLUME PATH** should point to the *new* copied and unzipped database, *not* the database from the original production environment.

4. It is possible to add multiple volumes using Quick Deploy. To add multiple volumes to the **Volume Information** table, add the following property to the **fnd0_tcdbsserver** component in the XML configuration file. The format for the volume information is:

```
<volume-name>;<original-host>;<copied-volume-path>;<volume-name>;<original-host>;<copied-  
volume-path>
```

For example:

```
<property id="fnd0_migratedVolumeInfo"  
value="volume1;vsc6s015;C:\apps\TC, volume2;vsc6s015;C:\apps\TC">
```

Database Creation Settings

☐ Create and populate database. Create new data directory.

For Oracle this means a table space will be created. For MSSQLServer a database will be created.

Database Path

☐ Populate database. Create new data directory.

A database exists, but is not populated with Teamcenter data. You want Deployment Center to populate the database and create a new TC DATA directory.

☒ Copy Environment using existing populated database.

The database is already populated. This option will create a copy of the environment and use this existing database.

Volume Information



VOLUME NAME	ORIGINAL HOST	COPIED VOLUME PATH
volume	vsc6s015	D:\apps\tc\TC\volume

- Update the environment site ID to the value of the original production system, which can be found in the Environment **Overview** page for your environment in Deployment Center, as well as the machine names where the new Teamcenter binaries will be placed with the existing database.
- If you are using Deployment Center 14.3.0.8 or later, navigate to **Overview** page for the test environment. Click **Start Edit** to edit the site ID.
- If you did not update the machine names in the XML configuration file when you imported the test environment, modify the machine names in the Deployment Center interface to match the machine names that will host the new Teamcenter binaries.

11. Update the preproduction environment

Perform update tasks on your preproduction Teamcenter environment to determine how you will update your production environment.

Prepare the pre-production environment for update

Install C Shell on Linux servers

Before upgrading a corporate server that is hosted on a Linux machine, you must verify that C Shell is installed.

If it is not installed, the Pre-Upgrade Diagnostic test fails at the Volume Access Check step with an error similar to the following:

```
Name: Volume Access Check
Description:
Test whether the FSC is up and running
Analysis:
An attempt to contact the FSC running at http://SERVER:4544 failed.
Check that the FSC is running and configured correctly.
```

If you see this error, install C Shell and rerun the Pre-Upgrade Diagnostic test.

Verify required groups and roles

Before you begin upgrade, verify the following groups exist in your database:

- **dba**
- **system**
- **Project Administration**
- **Validation Administration**

Before you begin upgrade, verify the following groups exist in your database:

- **dba**
- **system**

- **Project Administration**
- **Validation Administration**

Also, verify the following roles exist in the specified groups.

Role	Group
Project Administrator	Project Administration
Validation Administrator	Validation Administration

Upgrade may fail if these groups and roles are not present.

Verify required character set

To read and parse the XML files correctly, your system must be able to process non-English (non-ASCII) locale characters. To facilitate this, the system must be first loaded with the fonts for that locale. You must have the same locale installed on your Teamcenter host as you use to communicate with your database server, and the database server must support this locale as well.

On Linux systems, Teamcenter installation tools verify that the required character set is loaded by running the **locale -a** command in a shell. If the output does not list the required character set, you must add this character set before you upgrade Teamcenter.

Vorgehensweise

1. Log on to the system using the Common Desktop Environment (CDE) with the minimum required locale by choosing **Option→Language→<character-set>** during logon.

Alternatively, your system administrator may modify the date file (named **TIMEZONE** in the **etc** directory), which can preset this environment, so every time you log on and launch a shell, the environment is preset.
2. Set or export the **LC_ALL** environment variable by typing **LC_ALL=<character-set>** or the equivalent command for your platform.
3. Verify the setting using the **echo** command or equivalent. Make sure the correct value for **LC_ALL** is displayed.
4. Run the **locale** command and make sure the **LANG** variable and all the **LC_<x>** variables are set the same as **LC_ALL**.
5. If **LANG** is still set to **C**, manually export **LANG** to be the same value as **LC_ALL**.

If the required character set is not loaded on your machine, contact your system administrator to have it installed before you install the GM Overlay.

This is required because current Teamcenter versions use XML files rather than **.dat** files and associated scripts. Because of this, GM Overlay data is transformed from **.dat** files into XML files.

Update the pre-production environment

Upgrade Deployment Center

After the deployment of the test environment is successful, upgrade Deployment Center to Deployment Center 2606.

For more information, see *Upgrade Deployment Center in Deployment Center — Verwendung*.

Migrate BMIDE custom templates

After you upgrade Deployment Center to Deployment Center 2606, install the standalone Business Modeler IDE (BMIDE) and migrate custom software.

Vorgehensweise

1. Create a new environment that contains only the standalone Business Modeler IDE.
For more information, see *Install the Business Modeler IDE using Deployment Center - standalone in BMIDE für Datenmodelldesign*.
2. Use the BMIDE client to migrate the existing custom template to the new Teamcenter version and generate a software package.

Update the Software Repository

Vorgehensweise

1. Add the Teamcenter 2606 software kit:
 - a. Expand the Teamcenter 2606 software kit. Copy the unzipped contents to the *software* subdirectory in one of your registered repository locations.
 - b. In Deployment Center, click **SOFTWARE REPOSITORIES**.
The **Software Repositories** page opens the **Active Media** tab of the repository and displays the **Software Media** table.
 - c. Verify that the Teamcenter 2606 software appears in the list of available software. The list may take a few minutes to update.
If the software does not appear in the **Software Repositories** page, inspect the repository log files for repository scanning issues or software file problems. The repository log files are in the **webserver\repotool\logs** directory on the Deployment Center server.

2. Add the updated BMIDE package to the software repository.

Make sure the **Available Software** list includes the new Teamcenter software and your updated custom software.

Available Software

PRODUCT		SOFTWARE ▲
☒	 Teamcenter	Microservice Framework
☐	 Teamcenter	My Software
☒	 Teamcenter	My Software
☒	 Teamcenter	Teamcenter

Update the pre-production environment

After the test environment is successfully created, update the test environment to Teamcenter 2606 with the custom template, including generating and running deploy scripts. For more information, see *Update or patch your environment in Deployment Center — Verwendung*.

Review Deployment Center logs for errors or issues recorded during the update of the test environment. Logs are located in the **webserverlogs** directory on the Deployment Center machine.

Post-update tasks

Run the postinstallation tasks script (Linux systems)

On Linux systems, if you installed the corporate server without root privileges, a user with root privileges must run the root postinstallation tasks scripts. These scripts register daemons and perform other installation actions that require root privileges.

Run all scripts in the `<TC_ROOT>/install` directory that have names of the following form:

root_post_tasks<ID>.ksh

Replace `<ID>` with the unique part of each script name.

Enable a UTF-8 database

If you converted your Teamcenter database from non-UTF-8 to UTF-8 before you upgraded Teamcenter, update the **uTF8Enabled** setting in the **configuration.xml** file:

Vorgehensweise

1. Locate the `<TC_ROOT>\install\configuration.xml` file in your upgraded environment and open the file.
2. Locate the **uTF8Enabled** parameter in the file and set its value to **true**:

```
<uTF8Enabled value="true" />
```

3. Save the changes to the file.

Configure units of measure

The units of measure capability in Teamcenter changed in Teamcenter 2406 from the legacy Business Modeler IDE units of measure to the *unit management system* (UMS), which uses the **ums_mapping** utility. The utility links legacy units to unit definitions in the UMS.

If you are updating from a Teamcenter version earlier than Teamcenter 2406, you can migrate legacy units of measure to the unit management system as described in *Teamcenter Administration*.

Resolve Schedule Manager model errors

In current Teamcenter versions, the **Schedule**, **Schedule Task**, **Schedule Revision**, **Schedule Task Revision** and the associated forms (**Scheduling**, **Execution**, **SchMgtCostForm**) are re-parented from **Item** and **Item Revision** to **Workspace Object**.

In the default template, all references from **Schedule** and associated objects to **Item** and **Item Revision** have been corrected. However, for custom templates containing Schedule Manager customizations, the following steps must be performed to resolve any model errors during the Business Modeler IDE template migration.

Associated forms (**Scheduling**, **Execution**, **SchMgtCostForm**), **Schedule Revision** and **ScheduleTaskRevision** are not used by Schedule Manager and will be deprecated in a future release of Teamcenter.

Procedure

1. Examine model errors during migration of custom template to Teamcenter 2606

Study the model errors reported in the console of Business Modeler IDE during migration of the Business Modeler IDE custom template. These errors are typically in the **extensions/default.xml** and **extensions/lang/en_US/default_en_US.xml** files. Look for errors similar to the following examples:

- Property "pv5_MFK_Prop1" is not attached to Business Object "PV5_MFK_SchdTsk1" in TcTypeConstantAttach element.
- Invalid type found in constant attach element. "PV5_MFK_SchdTsk1" is not a subtype of any constant scope definition
- Incompatible MFK Definition: The property named "pv5_MFK_Prop1" on MFK definition "PV5_MFK_Require" is not valid on the business object "PV5_MFK_SchdTsk1".
- The OperationInputType "PV5_MFK_SchdTsk1RevisionRevI" cannot be loaded because the parent Type "ScheduleTaskRevisionRevI" does not exist.
- Property "pv5_MFK_Prop1" is not attached to Business Object "PV5_MFK_SchdTsk1" in TcPropertyConstantAttach element.
- Property "items_tag" is not attached to Business Object "Mr9a_SSJbTsk1Revision" in pathToSource element.
- Invalid type found in constant attach element. "Mr9a_SSJbTsk1" is not a subtype of any constant scope definition.
- Property "mr9a_jobact2prop" is not attached to Business Object "Mr9a_SSJbTsk1Revision" in Property{::}Mr9a_SSJbTsk1Revision{::}mr9a_jobact2prop element
- Attaching a Naming Rule to the "item_id" property on the "ScheduleTask" Business Object is not supported.

2. Resolve the model errors.

- a. Open the XML file in a text editor, XML editor, or the Business Modeler IDE text editor.
- b. Find the line number where the error is reported and delete the invalid element at that line number in the XML file.
- c. Save the XML file.
- d. Reload the custom template in BMIDE again and ensure there are no model errors being reported in the custom template.

Caution

Do not remove **TcStandardType** and **TcClass** model statements for any subtypes of **Schedule** and **ScheduleTask** business objects.

Results

After the upgrade process is complete, the custom **Schedule** and **ScheduleTask** business objects are automatically re-parented to **WorkspaceObject**.

Upgrade Zugriffs-Management rules

Special steps are required to upgrade the Zugriffs-Management rule tree in order to ensure the rule tree in your upgraded system contains any new rules added by Teamcenter and also any custom rules you added to your previous installation. You can upgrade rules in two ways. Choose the appropriate method depending on how many custom rules are in your Zugriffs-Management rule tree.

- If you have more custom rules than Teamcenter-supplied rules, migrate your legacy rule tree and then manually add the new Teamcenter-supplied rules:
 1. Create a backup of your existing rule tree using the Zugriffs-Management **Export** action.
 2. Upgrade your Teamcenter configuration to Teamcenter 2606.
During Teamcenter update, Deployment Center automatically imports your legacy rule tree to Teamcenter 2606.
 3. Identify changes in the rule tree by comparing the `..\<TC_DATA>\tcm_am_rule_tree.default` file in your previous environment to the same file in your Teamcenter 2606 environment.
 4. Start Zugriffs-Management and add rules introduced to Teamcenter since your previous version.
- If you have less custom rules than Teamcenter-supplied rules, use the standard Teamcenter 2606 rule tree and then manually add your custom rules.
 1. Create a backup of your existing rule tree using the Zugriffs-Management **Export** action.
 2. Identify your custom rules in order to add them after upgrade.
 3. Upgrade your Teamcenter configuration to Teamcenter 2606.
During Teamcenter update, Deployment Center automatically imports your legacy rule tree to Teamcenter 2606.
 4. Import the standard Teamcenter 2606 rule tree using the **am_install_tree** utility. Use the **mode=replace_all** argument to overwrite the legacy rule tree with the Teamcenter 2606 rule tree. The utility automatically creates ACLs and privileges during import.
 5. Manually add your custom rules into the rule tree in the appropriate locations.

Maintain default scope rules

Default scope rules (transfer modes, closure rules, filter rules, and so on) are maintained in XML files in the `<TC_DATA>` directory (for example, `defaultTransfermodes.xml`). The file is imported to Teamcenter using the **tcxml_import** command line utility during the database installation and upgrade.

The transfer mode `.xml` files are imported in overwrite mode as part of the upgrade. You need not manually import the files. To reload the latest changes from the `.xml` file, you can run the **tcxml_import**

utility in overwrite mode to ensure that the updates to existing scope rules are also imported to the database. For example:

```
$<TC_ROOT>/bin/tcxml_import -u=<user> -p=<password>  
-g=<group> -file=$<TC_DATA>/defaultTransfermodes.xml -scope_rules  
-scope_rules_mode=overwrite
```

Siemens Digital Industries Software recommends that you do not modify the default scope rules. If necessary, you can create a copy of the default rules and modify the copy as needed.

Achtung

All references to obsolete attributes and classes must be removed from custom closure rules or property sets. Additionally, remove any deprecated attributes and classes from custom closure rules or property sets.

For information about deprecated and obsolete classes and attributes, see the latest Teamcenter **README** file in the **Downloads** area on Support Center.

Set the time zone

Some Teamcenter applications including Schedule Manager and Workflow utilize calendars. When creating new calendars, Teamcenter determines the default time zone as follows:

1. Teamcenter checks the **Time Zone** property on the default calendar. If this property is set, its value is used.
2. If that property is not set, Teamcenter checks the **SiteTimeZone** preference. If this preference is set, its value is used.
3. If neither of these are set, Teamcenter uses GMT as the time zone.

If Teamcenter is using GMT and that is not the correct time zone for your site, you may encounter unexpected behavior. Therefore, Siemens Digital Industries Software recommends that you set both the **Time Zone** property on the default calendar using the Organization application and the **SiteTimeZone** preference.

Update Active Workspace components

Erneutes Zusammenführen von Solr- und Teamcenter-Schemas und Aktualisieren des Index

Nach dem Verschieben zu den Ziel-Active Workspace- und Ziel-Teamcenter-Versionen sind die Teamcenter-Solr-Schemas nicht synchronisiert. Sie können die Schemas zusammenführen und dann die Methode für

das Aktualisieren des Index auswählen. Sie müssen ermitteln, ob Sie einen vollständigen Index oder eine Delta-Neuindizierung Ihrer Daten benötigen, bevor Sie die Indexaktualisierung planen.

Bewerten des Deltas von Objektdatenänderungen

Wenn Ihre Indizierungsänderungen Hinzufügungen, Änderungen und Löschungen von Typen und Eigenschaften sind, können Sie eine Delta-Indizierungsaktualisierung anstatt einer vollständigen Neuindizierung durchführen.

Vorgehensweise

1. Zusammenführen der Schemata von Teamcenter und Solr.
2. Halten Sie die Synchronisierung durch den Indexer an, wenn sie ausgeführt wird.

```
runTcFTSIndexer -stop
```

3. Ermitteln Sie den Umfang der Änderungen zwischen dem letzten Indizierungsschema und dem aktuellen Schema.

Führen Sie das Dienstprogramm **awindexerutil** mit **-delta -dryrun** aus, um einen Bericht mit dem erwarteten Delta von Änderungen zu erhalten. Beispiel:

```
awindexerutil -u=<adminuser> -p=<password> -g=<group> -delta -dryrun
```

Die Unterschiede werden im Befehlsfenster und in einer Protokolldatei ausgegeben.

4. Beurteilen Sie nach dem Auswerten des Berichts, ob Sie das Delta von Änderungen für die Neuindizierung verwenden möchten. Wenn dies der Fall ist, führen Sie **awindexerutil** aus, um die Änderungen aus dem Bericht zu indizieren:

```
awindexerutil u=<adminuser> -p=<password> -g=<group> -delta
```

5. Testen Sie die Indexer-Konnektivität, indem Sie den Indexer-Testablauf ausführen.

```
runTcFTSIndexer -task=objdata:test
```

6. Starten Sie den Synchronisierungsablauf mit dem Dienstprogramm **runTcFTSIndexer**:

```
runTcFTSIndexer -task=objdata:sync -interval=<seconds>
```

Erneutes Indizieren Ihrer Daten bei Bedarf

In Ihrem Index sind möglicherweise viele Änderungen oder andere Arten von Änderungen vorhanden, die sich außerhalb des Umfangs von Delta-Änderungen befinden. Wenn dies der Fall ist, indizieren Sie stattdessen Ihre Daten erneut:

Vorgehensweise

1. Zusammenführen der Schemata von Teamcenter und Solr.
2. Testen Sie die Indexer-Konnektivität, indem Sie den Indexer-Testablauf ausführen.

```
runTcFTSIndexer -task=objdata:test
```

3. Indizieren Sie die Daten neu.

Additional Active Workspace postupdate steps

Import table column configuration

If you have configured any custom table columns other than those available by default, perform the following after upgrading or patching:

Procedure

1. Update the custom column information in the **Awb0ContentTableUiConfigCots.xml** file.
2. Run the **import_uiconfig** utility to import the column configuration.

Configure content security policy directives

If you created content security policy (CSP) directives in your previous Teamcenter environment and recorded them before updating Teamcenter, re-enter your CSP directives in the **Active Workspace Gateway** component as described in Content security policy directive configuration in *Teamcenter-Sicherheit*.

Reindex classification data

If you are new to classification and want to search for classification data in Active Workspace using classification standard taxonomy, your first step is to create a classification hierarchy.

If you already use traditional classification and want to search for classification data in Active Workspace or Teamcenter Manufacturing Access, perform the following steps:

Procedure

1. Create search index views and specify facetable properties.
2. Index classification data.
3. Index or reindex your data. If your data is already indexed, it is sufficient to index only the modified classification data.

Set up a custom template after patching

If you have a custom template that is dependent on the Active Workspace template (**aws2**) and other Active Workspace templates, you must follow a special procedure to set up this template after patching Active Workspace.

Procedure

1. Add the Active Workspace template (**aws2**) any other needed Active Workspace templates to the Business Modeler IDE.
2. In Deployment Center, generate deploy scripts and run them on affected machines. This copies the required Business Modeler IDE .jar files for Active Workspace into the installation.
3. In the Business Modeler IDE, import your custom template. To add the Active Workspace template (**aws2**) and the other dependent Active Workspace templates:
 - a. Right-click your custom template project and choose **Properties**.
 - b. In the left pane, choose **Teamcenter→BMIDE**.
 - c. In the **Dependent Templates** pane, select the **aws2** template and the other Active Workspace templates previously added using the **Add/Update Templates for working within the Business Modeler IDE Client** option in TEM.
 - d. Click **OK**.
The template is now ready to be deployed.

Restart Teamcenter Process Manager

After completing updates to Active Workspace, ensure the updates made to the Teamcenter GraphQL Service take effect:

Procedure

1. Stop the Teamcenter Process Manager service in the **Services** dialog of the Windows Control Panel.
2. Wait for all affected processes to stop, at least 15 seconds.
3. Restart the Teamcenter Process Manager service.

Results

Some Active Workspace features such as Discussions (Active Collaboration) and Reports rely on updates to the Teamcenter GraphQL Service in order to function properly. This restart process ensures the updates are implemented for those features.

Additional postupgrade steps

- Transfer access rules for Classification Admin objects

If you created classification access rules for admin objects that resided in the **Has Class(POM_object)** area, you must move these rules manually to the **Has Application(Any)** area of the rule tree after upgrade.

In current versions of Teamcenter, access to classification administrative objects is restricted to dba users, but you can grant other users access to classification objects by creating a rule for each of these users or groups of users in the **Has Application(Any)** group in the rule tree.

- Add new Klassifizierung units

When you upgrade Teamcenter, new Klassifizierung units are not added because they may conflict with other units created by Teamcenter administrators. Update unit definitions manually as described in *Grundlegende Klassifizierung — Bereitstellung und Administration*.

- Set variant rule preference

If you already use classic variants, you can use your existing variability data with the Produkt-Konfigurator. You can now use variant rules to store variant configuration criteria and optional validation records. They are attached to the product item revision or the application model (collaborative design) by GRM relationships.

Variant rules perform the same function as stored option sets (SOSs), which will be deprecated in a future version. Siemens Digital Industries Software encourages customers who use classic variants to transition to variant rules by setting the **DisableSaveSos** preference to **TRUE**. This setting prevents users from saving SOSs, thereby forcing them to create variant rules.

Existing modular variant customers should not change the default preference setting of **false**.

- Transfer HTTPS certificates

If you use HTTPS transports, you must transfer all the HTTPS certificate information to the upgraded Teamcenter installation.

14. Deploy the update to your production environment

Apply the update process you developed on your pre-production Teamcenter environment to your production environment.

Update volume servers

If you have volume servers (Teamcenter configurations that contain an FSC separate from the corporate server), you must update those configurations *before* you update the corporate server.

This is because Teamcenter utilities that run during the corporate server update are not compatible with FSCs from other Teamcenter versions. Deployment Center updates the FSC for the corporate server during update of the corporate server.

Update the production environment

Update your production environment using Deployment Center.

Achtung

Make sure you **back up your Teamcenter data** before you update your production environment.

After you up verify that you successfully updated your test environment and you have resolved any issues found during the test update, update your production environment using the same procedure you used to update your test environment. Generate and run deploy scripts to complete the update.

For more information, see *Update or patch your environment* in *Deployment Center — Verwendung*.

16. Complete the update

Perform steps to complete the update of your Teamcenter production environment, and perform optional additional steps to maintain and enhance your environment.

Apply updates to your environment

Deployment Center software updates follow a process that is similar to installation. When you want to update your environment, choose the target version of software you want to apply, for example, Teamcenter 2606.0001. Deployment Center determines what to update based on what is required by the selected target release and selected applications.

Before you perform an update, you must download, unzip, and put the software update or patch kits in the Deployment Center repository. Check the repository for software dependencies and messages about missing software.

Import a configuration from TEM

Deployment Center can update or patch software in a registered Teamcenter environment. If the environment you want to update originated in TEM, run the **send_configuration_to_dc** script on the target servers to send the latest environment configuration information to Deployment Center.

You may not be required to put source release software kits in the repository. Deployment Center constructs source to target release mapping using the environment's current configuration files sent in the report from *send_configuration_to_dc*. Deployment Center analyzes the target release information to construct the mapping at the time you choose to update. Sometimes, Deployment Center may require the source software if the target release doesn't provide adequate mapping information. If there is missing software that is required, Deployment Center displays messages telling you about the dependency and how to proceed.

Perform updates or patches

The procedure for updating an environment is similar to creating an environment.

Vorgehensweise

1. Open the **Environments** page and choose the environment where you want to update software. Choose **Deploy Software** to begin the process.
2. **Software**

Choose the target update software from the **Available Software** list. The **Selected Software** list displays currently installed versions and latest pending versions for the environment. If the software you need is not available, check whether it was listed in the repository.

If missing software is required, Deployment Center tells you about the dependency and how to proceed.

3. Options

Either Single Box or Distributed is selected for **Environment Type**. If you previously had a **Single Box** environment, you can choose **Distributed**; however, you will need to update the server information for components.

If an environment is already deployed on multiple servers, **Single Box** is not available.

The **Architecture Type** for your environment is automatically selected and can't be changed.

4. Applications

Applications that are already installed are automatically included for update. You can add other applications from the list. Applications that display a **Pending Install** status are waiting for deployment. Applications that are installed but need updates display the **Pending Update** status.

See *Application names changed in Deployment Center* for additional information about application names.

5. Components

Components that are not yet installed display the **Pending Install** status. Components that are installed but need updates to support your selected applications display the **Pending Update** status.

It is possible that a selection from the current update may cause a previously configured component to need more information.

Components display the % configured. If it's less than **100%**, complete the required parameter values. Components that are not impacted can be ignored (showing **100%** configured).

6. Deploy

Generate deployment scripts for the update. This tab is available when the **Components** tab is complete.

Deployment scripts contain the information you configured in Deployment Center for the selected environment.

7. Run the deployment scripts.

After the scripts and software ZIP files are generated, copy them to each target machine and run them.

17. Appendices

Supplemental procedures and references for updating Teamcenter and Active Workspace.

Verwandte Themen

[Teamcenter Administration](#)

[BMIDE for Data Model Design](#)

[Teamcenter Rich Client Installation on Windows Using TEM](#)

[Teamcenter Rich Client Installation on Linux Using TEM](#)

Troubleshooting

Troubleshooting Microsoft SQL Server

Microsoft SQL Server 2014 performance is poor

If you migrate a database application to Microsoft SQL Server 2014 from a previous version, the database server may consume excessive CPU resources and cause poor performance.

To correct this problem, change the SQL Server 2014 Compatibility Level setting from SQL Server 2014 (120) to SQL Server 2012 (110).

For more information about this issue, see the following Microsoft support article:

<https://msdn.microsoft.com>

Teamcenter update fails with ODBC error

When upgrading a Microsoft SQL Server server, an error similar to the following can occur:

```
+++++
ODBC error. SQLSTATE: 42000 Native error: 5074
Message: [Microsoft][ODBC SQL Server Driver][SQL Server]The column '****' is
dependent on column '****'.
ODBC error. SQLSTATE: 42000 Native error: 4922
Message: [Microsoft][ODBC SQL Server Driver][SQL Server]ALTER TABLE ALTER
COLUMN
<name> failed because one or more objects access this column.
+++++
```

This error occurs when the upgrade process attempts to modify a column that has a dependent column with an index. Microsoft SQL Server does not allow changes to columns with indexes. Also, local DBA indexes may exist that don't match the standard OOTB template for indexes, so it was not anticipated.

This problem can happen because columns that have manually-created statistics attached cannot have their properties modified without first dropping the statistics object. This to ensure the statistics object accurately reflects the content of the column. Manual creation of statistics objects is important to ensuring query performance if you set `AUTO_CREATE_STATISTICS = OFF`.

An auto-created statistics object does not prevent a modify action to a column because auto-created statistics objects can be removed automatically. But, if the system encounters a manually-created statistics object, it cannot be removed automatically, and may result in an access error.

To resolve this problem, perform the following steps:

Vorgehensweise

- 1. Delete the index ***.
- 2. Delete the dependent column ***.
- 3. Continue the upgrade.
- 4. Run the **index_verifier** utility to re-create standard OOTB indexes:

```
index_verifier -u=<tc-admin> -p= -g=dba -o=DO_IT
```

Replace <tc-admin> with the Teamcenter administrative user.

Troubleshooting Teamcenter upgrade

Possible solutions to upgrade issues

Locate possible solutions to problems you may encounter during Teamcenter upgrade.

Problem	Possible cause	Solution
Errors when uploading or downloading files	FMS_HOME environment variable does not point to the location of the Dateiverwaltungssystem (FMS) client executables	Uninstall the Rich Client as described in the appropriate client installation instructions (for Windows or Linux). Then reinstall the Rich Client.
	Incomplete installation of FMS client cache executables	Achtung Always uninstall a Rich Client using this procedure before installing a new Rich Client.
	Incompatible configurations of FMS	If two Rich Clients are installed on the workstation, report the errors to the Teamcenter administrator. When two Rich Clients are installed on the same workstation, FMS uses the FMS_HOME value set during the first installation. The configuration of

Problem	Possible cause	Solution
Teamcenter displays an error message when you attempt to access Teamcenter online help	The Teamcenter administrator did not include online help when configuring your Rich Client	the file client cache (FCC) to a file server cache (FSC) must support both deployments, especially if the second deployment is to a different database. Contact the Teamcenter administrator.
Unable to view or to search Teamcenter online help	The Web browser you are using does not meet requirements to view and search Teamcenter online help	Make sure you are using a Siemens Digital Industries Software-supported Web browser. For information about supported Web browsers, see the Siemens Digital Industries Software Certification Database .
Teamcenter upgrade fails with custom types	A custom data type has the same name as the parent class.	If you have a custom type name with the same name as the parent class, the custom type is identified and added to the custom templates, but the upgrade fails. For example, you have a custom storage_media type that belongs to the storage_media class. Before upgrading Teamcenter, rename the custom type name to anything other than the parent class. For example, rename the storage_media type to storage_media_custom before you upgrade.
Teamcenter upgrade fails with large database issues	The database is not sized properly.	If your upgrade changes or deletes a large number of records, it can fail if the database is not sized properly. Oracle displays an error similar to the following:

```
ORA-01652: unable to extend temp
segment by 128 in
tablespace TEMP
```

The following upgrade examples can lead to this error:

```
install -mod_attr <tc-admin> $
{TC_USER_PASSWD}
  dba POM_application_object
owning_user
  POM_attr_no_pom_backpointer +
```

```
install -mod_attr <tc-admin> $
{TC_USER_PASSWD}
  dba POM_application_object
last_mod_user
  POM_attr_no_pom_backpointer +
```

Replace **<tc-admin>** with the Teamcenter administrative user.

These commands update many rows in the **pom_backpointer** table. The number of rows updated is approximately three times the size of the **ppom_application_object** table. You can determine the table size with the following SQL command:

```
SELECT COUNT(puid) FROM
ppom_application_object;
```

Problem	Possible cause	Solution
---------	----------------	----------

To resolve this issue, you must have a large undo or rollback tablespace and a large **TEMP** tablespace. Increase the size of both the **TEMP** and undo tablespaces to at least 5 GB each.

1. Use the following SQL command to determine where the data files reside:

```
SELECT tablespace_name, file_name
FROM dba_data_files;
```

The results may resemble the following:

TABLESPACE_NAME	FILE_NAME
SYSTEM	D:\ORACLE\ORADATA\SYSTEM01.DBF
UNDOTBS1	D:\ORACLE\ORADATA\UNDOTBS01.DBF
SYSAUX	D:\ORACLE\ORADATA\SYSAUX01.DBF
IDATA	D:\ORACLE\ORADATA\IDATA01.DBF
ILOG	D:\ORACLE\ORADATA\ILOG01.DBF
INDX	D:\ORACLE\ORADATA\INDX01.DBF

In this case, all data files are located in the D:\ORACLE\ORADATA directory.

2. Add a data file to the **TEMP** tablespace:

```
ALTER TABLESPACE TEMP ADD
tempfile
D:\ORACLE\ORADATA\temp2.dbf
size 5000M;
```

3. Add a data file to the undo tablespace:

```
ALTER TABLESPACE UNDOTBS01 ADD
datafile
'D:\ORACLE\ORADATA\
undotbs2.dbf' size 5000M;
```

To help ensure the upgrade is successful, make sure the redo logs are sized properly (100M each) and reside on a separate disk from the data. If necessary, resize the redo logs.

1. Set the **ORACLE_SID** environment variable.
2. Log on to Oracle:

```
sqlplus /nolog
connect system/<password> as
sysdba;
```

Problem	Possible cause	Solution
---------	----------------	----------

- Find the list of the redo log groups and see where the files reside.

```
SELECT GROUP#, TYPE, STATUS, MEMBER
FROM V$LOGFILE;
```

The results may look like this:

```
3 ONLINE STALE
E:\ORACLE\ORADATA\DPVPERF\RED003.
LOG
2 ONLINE
E:\ORACLE\ORADATA\DPVPERF\RED002.
LOG
1 ONLINE STALE
E:\ORACLE\ORADATA\DPVPERF\RED001.
LOG
```

- Create the new redo logs as follows. You can change the drive where you can create these, but redo logs should reside on a separate disk from the data.

```
ALTER DATABASE ADD LOGFILE GROUP
4
('E:\ORACLE\ORADATA\DPVPERF\
RED004.LOG') SIZE 100M;
ALTER DATABASE ADD LOGFILE GROUP
5
('E:\ORACLE\ORADATA\DPVPERF\
RED005.LOG') SIZE 100M;
ALTER DATABASE ADD LOGFILE GROUP
6
('E:\ORACLE\ORADATA\DPVPERF\
RED006.LOG') SIZE 100M;
```

- Run the following command to force the log switch:

```
ALTER SYSTEM SWITCH LOGFILE;
```

- (Optional) Drop redo log groups 1, 2, and 3:

```
ALTER DATABASE DROP LOGFILE
GROUP 1;
ALTER DATABASE DROP LOGFILE
GROUP 2;
ALTER DATABASE DROP LOGFILE
GROUP 3;
```

Problem	Possible cause	Solution
		Anmerkung If the group number is the current group and you cannot drop it, run the command to force the log switch to make group 4, 5, or 6 current.
Rich Client startup performance is slow	Wait time for hierarchy tree checked-out icon display.	Set the TC_show_checkedout_icon preference to FALSE. Anmerkung Users and administrators can set the TC_show_checkedout_icon preference to TRUE to enhance usability or to FALSE to enhance Rich Client startup performance. For more information, see the <i>Teamcenter-Umgebungsvariablen</i>.